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Inventor(s)	Natarajan; Vivek et al.

Processing multimodal user input for assistant systems

Abstract

In one embodiment, a method includes receiving at a head-mounted device a speech input from a user and a visual input captured by cameras of the head-mounted device, wherein the visual input comprises subjects and attributes associated with the subjects, and wherein the speech input comprises a co-reference to one or more of the subjects, resolving entities corresponding to the subjects associated with the co-reference based on the attributes and the co-reference, and presenting a communication content responsive to the speech input and the visual input at the head-mounted device, wherein the communication content comprises information associated with executing results of tasks corresponding to the resolved entities.

Inventors: **Natarajan; Vivek (Sunnyvale, CA), Mei; Shawn C. P. (San Francisco, CA), Zuo; Zhengping (Medina, WA)**

Applicant: **Meta Platforms, Inc.** (Menlo Park, CA)

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Assignee: **Meta Platforms, Inc. (Menlo Park, CA)**

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2012/0109858	12/2011	Makadia et al.	N/A	N/A
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2012/0159635	12/2011	He et al.	N/A	N/A
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Primary Examiner: Le; Hung D

Attorney, Agent or Firm: Morgan, Lewis & Bockius LLP

Background/Summary

PRIORITY (1) This application is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 17/156,964, filed 25 Jan. 2021, which is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 16/053,600, filed 2 Aug. 2018, which claims the benefit, under 35 U.S.C. § 119(e), of U.S. Provisional Patent Application No. 62/660,876, filed 20 Apr. 2018, each of which is incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure generally relates to databases and file management within network environments, and in particular relates to hardware and software for smart assistant systems.

BACKGROUND

(2) An assistant system can provide information or services on behalf of a user based on a combination of user input, location awareness, and the ability to access information from a variety of online sources (such as weather conditions, traffic congestion, news, stock prices, user schedules, retail prices, etc.). The user input may include text (e.g., online chat), especially in an instant messaging application or other applications, voice, images, or a combination of them. The assistant system may perform concierge-type services (e.g., making dinner reservations, purchasing event tickets, making travel arrangements) or provide information based on the user input. The assistant system may also perform management or data-handling tasks based on online information and events without user initiation or interaction. Examples of those tasks that may be performed by an assistant system may include schedule management (e.g., sending an alert to a dinner date that a user is running late due to traffic conditions, update schedules for both parties, and change the restaurant reservation time). The assistant system may be enabled by the combination of computing devices, application programming interfaces (APIs), and the proliferation of applications on user devices.

(3) A social-networking system, which may include a social-networking website, may enable its users (such as persons or organizations) to interact with it and with each other through it. The social-networking system may, with input from a user, create and store in the social-networking system a user profile associated with the user. The user profile may include demographic information, communication-channel information, and information on personal interests of the user. The social-networking system may also, with input from a user, create and store a record of relationships of the user with other users of the social-networking system, as well as provide services (e.g. profile/news feed posts, photo-sharing, event organization, messaging, games, or advertisements) to facilitate social interaction between or among users.

(4) The social-networking system may send over one or more networks content or messages related to its services to a mobile or other computing device of a user. A user may also install software applications on a mobile or other computing device of the user for accessing a user profile of the user and other data within the social-networking system. The social-networking system may generate a personalized set of content objects to display to a user, such as a newsfeed of aggregated stories of other users connected to the user.

SUMMARY OF PARTICULAR EMBODIMENTS

(5) In particular embodiments, the assistant system may assist a user to obtain information or services. The assistant system may enable the user to interact with it with multi-modal user input (such as voice, text, image, video) in stateful and multi-turn conversations to get assistance. The assistant system may create and store a user profile comprising both personal and contextual information associated with the user. In particular embodiments, the assistant system may analyze

the user input using natural-language understanding. The analysis may be based on the user profile for more personalized and context-aware understanding. The assistant system may resolve entities associated with the user input based on the analysis. In particular embodiments, the assistant system may interact with different agents to obtain information or services that are associated with the resolved entities. The assistant system may generate a response for the user regarding the information or services by using natural-language generation. Through the interaction with the user, the assistant system may use dialog management techniques to manage and forward the conversation flow with the user. In particular embodiments, the assistant system may further assist the user to effectively and efficiently digest the obtained information by summarizing the information. The assistant system may also assist the user to be more engaging with an online social network by providing tools that help the user interact with the online social network (e.g., creating posts, comments, messages). The assistant system may additionally assist the user to manage different tasks such as keeping track of events. In particular embodiments, the assistant system may proactively execute tasks that are relevant to user interests and preferences based on the user profile without a user input. In particular embodiments, the assistant system may check privacy settings to ensure that accessing a user's profile or other user information and executing different tasks are permitted subject to the user's privacy settings.

(6) In particular embodiments, the assistant system may receive a multimodal user input (e.g., sound, image, video, text) from a client system associated with a user. The assistant system may process the multimodal user input with different modules of the assistant system based on the different modalities of the user input. The assistant system may then understand the user input, i.e., identify subjects/entities and semantic meaning of the user input, based on an entity resolution module and a co-reference module. The assistant system may further generate a multimodal output based on the understanding and send the multimodal output to the user via the client system. As an example and not by way of limitation, a user may submit a picture of several people to the assistant system and ask the assistant system to initiate a call to one of them (e.g., the user may say “call the guy on the left”). The assistant system may determine the identities of these people, identify the person “on the left,” access the contact information of the person on the left (subject to that person's privacy settings), and initiate a call for the user accordingly. In particular, the assistant system may use a visual-recognition agent capable of processing visual input included in the multimodal user input, where the visual-recognition agent may access various computer-vision models to understand the visual input. The understanding of the visual input may be used by the co-reference module to resolve the entities associated with the user input. Based on the understanding of the visual input and the resolved entities, the assistant system may use an assistant xbot to conduct a conversation with the user to further enhance the user's experience with the assistant system by executing different tasks corresponding to the visual input. The assistant system may reactively process multimodal user input in response to a user query. The assistant system may also proactively process a multimodal user input by making suggestions based on the user input. The capability of processing multimodal user input and seamlessly switching between different modalities of output makes the assistant system well adapted to various client systems and software, thus making it more useful and appealing to users. Although this disclosure describes processing particular multimodal user input via a particular system in a particular manner, this disclosure contemplates processing any suitable multimodal user input via any suitable system in any suitable manner.

(7) In particular embodiments, the assistant system may receive, from a client system associated with a first user, a user input based on one or more modalities. At least one of the modalities of the user input may be a visual modality. In particular embodiments, the assistant system may identify, based on one or more machine-learning models, one or more subjects associated with the user input based on the visual modality. In particular embodiments, the assistant system may determine, based on the one or more machine-learning models, one or more attributes associated with the one or

more subjects, respectively. The assistant system may then resolve, based on the determined one or more attributes, one or more entities corresponding to the one or more subjects. In particular embodiments, the assistant system may execute one or more tasks associated with the one or more resolved entities. The assistant system may further send, to the client system associated with the first user, instructions for presenting a communication content responsive to user input. The communication content may comprise information associated with the executed one or more tasks.

(8) Certain technical challenges exist for achieving the goal of processing and understanding multimodal user input. One technical challenge includes accurately identifying, from visual input, subjects and their attributes. The solutions presented by the embodiments disclosed herein to address the above challenge are machine-learning models based on facial recognition and object detection, which are effective for recognizing people, locations, businesses, and objects together with their attributes. Another technical challenge includes accurately resolving entities from multimodal user input. The solution presented by the embodiments disclosed herein to address this challenge is a co-reference module in the assistant system, which is able to link the wording information of a voice and/or textual input with the visual analysis result of a visual input, thereby accurately resolving entities of the multimodal user input. Another technical challenge includes responding to multimodal user input with communication contents that are based on suitable modalities. The solution presented by the embodiments disclosed herein to address this challenge is determining proper modalities based on contextual information associated with both a user and a client system associated with the user, which allows the assistant system to communicate with the user based on the determined modalities that may be more suitable for the user's current situation.

(9) Certain embodiments disclosed herein may provide one or more technical advantages. A technical advantage of the embodiments may include efficiently adapting to various client systems and software with the ability of processing multimodal user input and switching between different modalities of output. Another technical advantage of the embodiments may include improved understanding of multimodal user input as the voice and/or textual input and visual input are analyzed jointly via a co-reference module. Another technical advantage of the embodiments may include increasing the degree of users engaging with the assistant system by proactively suggesting tasks to users based on analysis of the visual input. Certain embodiments disclosed herein may provide none, some, or all of the above technical advantages. One or more other technical advantages may be readily apparent to one skilled in the art in view of the figures, descriptions, and claims of the present disclosure.

(10) The embodiments disclosed herein are only examples, and the scope of this disclosure is not limited to them. Particular embodiments may include all, some, or none of the components, elements, features, functions, operations, or steps of the embodiments disclosed herein. Embodiments according to the invention are in particular disclosed in the attached claims directed to a method, a storage medium, a system and a computer program product, wherein any feature mentioned in one claim category, e.g. method, can be claimed in another claim category, e.g. system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates an example network environment associated with an assistant system.

(2) FIG. **2** illustrates an example architecture of the assistant system.

(3) FIG. **3** illustrates an example diagram flow of responding to a user request by the assistant system.

(4) FIG. **4** illustrates an example diagram flow of processing multimodal user input based on the example architecture of the assistant system in FIG. **2**.

(5) FIG. **5** illustrates an example interaction between a user and the assistant system via multimodal user input and system output.

(6) FIG. **6** illustrates an example method for processing multimodal user input.

(7) FIG. **7** illustrates an example social graph.

(8) FIG. **8** illustrates an example view of an embedding space.

(9) FIG. **9** illustrates an example artificial neural network.

(10) FIG. **10** illustrates an example computer system.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(11) System Overview

(12) FIG. **1** illustrates an example network environment **100** associated with an assistant system. Network environment **100** includes a client system **130**, an assistant system **140**, a social-networking system **160**, and a third-party system **170** connected to each other by a network **110**. Although FIG. **1** illustrates a particular arrangement of a client system **130**, an assistant system **140**, a social-networking system **160**, a third-party system **170**, and a network **110**, this disclosure contemplates any suitable arrangement of a client system **130**, an assistant system **140**, a social-networking system **160**, a third-party system **170**, and a network **110**. As an example and not by way of limitation, two or more of a client system **130**, a social-networking system **160**, an assistant system **140**, and a third-party system **170** may be connected to each other directly, bypassing a network **110**. As another example, two or more of a client system **130**, an assistant system **140**, a social-networking system **160**, and a third-party system **170** may be physically or logically co-located with each other in whole or in part. Moreover, although FIG. **1** illustrates a particular number of client systems **130**, assistant systems **140**, social-networking systems **160**, third-party systems **170**, and networks **110**, this disclosure contemplates any suitable number of client systems **130**, assistant systems **140**, social-networking systems **160**, third-party systems **170**, and networks **110**. As an example and not by way of limitation, network environment **100** may include multiple client systems **130**, assistant systems **140**, social-networking systems **160**, third-party systems **170**, and networks **110**.

(13) This disclosure contemplates any suitable network **110**. As an example and not by way of limitation, one or more portions of a network **110** may include an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a cellular telephone network, or a combination of two or more of these. A network **110** may include one or more networks **110**.

(14) Links **150** may connect a client system **130**, an assistant system **140**, a social-networking system **160**, and a third-party system **170** to a communication network **110** or to each other. This disclosure contemplates any suitable links **150**. In particular embodiments, one or more links **150** include one or more wireline (such as for example Digital Subscriber Line (DSL) or Data Over Cable Service Interface Specification (DOCSIS)), wireless (such as for example Wi-Fi or Worldwide Interoperability for Microwave Access (WiMAX)), or optical (such as for example

Synchronous Optical Network (SONET) or Synchronous Digital Hierarchy (SDH)) links. In particular embodiments, one or more links **150** each include an ad hoc network, an intranet, an extranet, a VPN, a LAN, a WLAN, a WAN, a WWAN, a MAN, a portion of the Internet, a portion of the PSTN, a cellular technology-based network, a satellite communications technology-based network, another link **150**, or a combination of two or more such links **150**. Links **150** need not necessarily be the same throughout a network environment **100**. One or more first links **150** may differ in one or more respects from one or more second links **150**.

(15) In particular embodiments, a client system **130** may be an electronic device including hardware, software, or embedded logic components or a combination of two or more such components and capable of carrying out the appropriate functionalities implemented or supported by a client system **130**. As an example and not by way of limitation, a client system **130** may include a computer system such as a desktop computer, notebook or laptop computer, netbook, a tablet computer, e-book reader, GPS device, camera, personal digital assistant (PDA), handheld electronic device, cellular telephone, smartphone, smart speaker, other suitable electronic device, or any suitable combination thereof. In particular embodiments, the client system **130** may be a smart assistant device. More information on smart assistant devices may be found in U.S. patent application Ser. No. 15/949,011, filed 9 Apr. 2018, U.S. Patent Application No. 62/655,751, filed 10 Apr. 2018, U.S. patent application Ser. No. 29/631,910, filed 3 Jan. 2018, U.S. patent application Ser. No. 29/631,747, filed 2 Jan. 2018, U.S. patent application Ser. No. 29/631,913, filed 3 Jan. 2018, and U.S. patent application Ser. No. 29/631,914, filed 3 Jan. 2018, which are incorporated by reference. This disclosure contemplates any suitable client systems **130**. A client system **130** may enable a network user at a client system **130** to access a network **110**. A client system **130** may enable its user to communicate with other users at other client systems **130**.

(16) In particular embodiments, a client system **130** may include a web browser **132** and may have one or more add-ons, plug-ins, or other extensions. A user at a client system **130** may enter a Uniform Resource Locator (URL) or other address directing a web browser **132** to a particular server (such as server **162**, or a server associated with a third-party system **170**), and the web browser **132** may generate a Hyper Text Transfer Protocol (HTTP) request and communicate the HTTP request to server. The server may accept the HTTP request and communicate to a client system **130** one or more Hyper Text Markup Language (HTML) files responsive to the HTTP request. The client system **130** may render a web interface (e.g. a webpage) based on the HTML files from the server for presentation to the user. This disclosure contemplates any suitable source files. As an example and not by way of limitation, a web interface may be rendered from HTML files, Extensible Hyper Text Markup Language (XHTML) files, or Extensible Markup Language (XML) files, according to particular needs. Such interfaces may also execute scripts, combinations of markup language and scripts, and the like. Herein, reference to a web interface encompasses one or more corresponding source files (which a browser may use to render the web interface) and vice versa, where appropriate.

(17) In particular embodiments, a client system **130** may include a social-networking application **134** installed on the client system **130**. A user at a client system **130** may use the social-networking application **134** to access on online social network. The user at the client system **130** may use the social-networking application **134** to communicate with the user's social connections (e.g., friends, followers, followed accounts, contacts, etc.). The user at the client system **130** may also use the social-networking application **134** to interact with a plurality of content objects (e.g., posts, news articles, ephemeral content, etc.) on the online social network. As an example and not by way of limitation, the user may browse trending topics and breaking news using the social-networking application **134**.

(18) In particular embodiments, a client system **130** may include an assistant application **136**. A user at a client system **130** may use the assistant application **136** to interact with the assistant system **140**. In particular embodiments, the assistant application **136** may comprise a stand-alone

application. In particular embodiments, the assistant application **136** may be integrated into the social-networking application **134** or another suitable application (e.g., a messaging application). In particular embodiments, the assistant application **136** may be also integrated into the client system **130**, an assistant hardware device, or any other suitable hardware devices. In particular embodiments, the assistant application **136** may be accessed via the web browser **132**. In particular embodiments, the user may provide input via different modalities. As an example and not by way of limitation, the modalities may include audio, text, image, video, etc. The assistant application **136** may communicate the user input to the assistant system **140**. Based on the user input, the assistant system **140** may generate responses. The assistant system **140** may send the generated responses to the assistant application **136**. The assistant application **136** may then present the responses to the user at the client system **130**. The presented responses may be based on different modalities such as audio, text, image, and video. As an example and not by way of limitation, the user may verbally ask the assistant application **136** about the traffic information (i.e., via an audio modality). The assistant application **136** may then communicate the request to the assistant system **140**. The assistant system **140** may accordingly generate the result and send it back to the assistant application **136**. The assistant application **136** may further present the result to the user in text.

(19) In particular embodiments, an assistant system **140** may assist users to retrieve information from different sources. The assistant system **140** may also assist user to request services from different service providers. In particular embodiments, the assist system **140** may receive a user request for information or services via the assistant application **136** in the client system **130**. The assist system **140** may use natural-language understanding to analyze the user request based on user's profile and other relevant information. The result of the analysis may comprise different entities associated with an online social network. The assistant system **140** may then retrieve information or request services associated with these entities. In particular embodiments, the assistant system **140** may interact with the social-networking system **160** and/or third-party system **170** when retrieving information or requesting services for the user. In particular embodiments, the assistant system **140** may generate a personalized communication content for the user using natural-language generating techniques. The personalized communication content may comprise, for example, the retrieved information or the status of the requested services. In particular embodiments, the assistant system **140** may enable the user to interact with it regarding the information or services in a stateful and multi-turn conversation by using dialog-management techniques. The functionality of the assistant system **140** is described in more detail in the discussion of FIG. **2** below.

(20) In particular embodiments, the social-networking system **160** may be a network-addressable computing system that can host an online social network. The social-networking system **160** may generate, store, receive, and send social-networking data, such as, for example, user-profile data, concept-profile data, social-graph information, or other suitable data related to the online social network. The social-networking system **160** may be accessed by the other components of network environment **100** either directly or via a network **110**. As an example and not by way of limitation, a client system **130** may access the social-networking system **160** using a web browser **132**, or a native application associated with the social-networking system **160** (e.g., a mobile social-networking application, a messaging application, another suitable application, or any combination thereof) either directly or via a network **110**. In particular embodiments, the social-networking system **160** may include one or more servers **162**. Each server **162** may be a unitary server or a distributed server spanning multiple computers or multiple datacenters. Servers **162** may be of various types, such as, for example and without limitation, web server, news server, mail server, message server, advertising server, file server, application server, exchange server, database server, proxy server, another server suitable for performing functions or processes described herein, or any combination thereof. In particular embodiments, each server **162** may include hardware, software, or embedded logic components or a combination of two or more such components for carrying out

the appropriate functionalities implemented or supported by server **162**. In particular embodiments, the social-networking system **160** may include one or more data stores **164**. Data stores **164** may be used to store various types of information. In particular embodiments, the information stored in data stores **164** may be organized according to specific data structures. In particular embodiments, each data store **164** may be a relational, columnar, correlation, or other suitable database. Although this disclosure describes or illustrates particular types of databases, this disclosure contemplates any suitable types of databases. Particular embodiments may provide interfaces that enable a client system **130**, a social-networking system **160**, or a third-party system **170** to manage, retrieve, modify, add, or delete, the information stored in data store **164**.
(21) In particular embodiments, the social-networking system **160** may store one or more social graphs in one or more data stores **164**. In particular embodiments, a social graph may include multiple nodes-which may include multiple user nodes (each corresponding to a particular user) or multiple concept nodes (each corresponding to a particular concept)—and multiple edges connecting the nodes. The social-networking system **160** may provide users of the online social network the ability to communicate and interact with other users. In particular embodiments, users may join the online social network via the social-networking system **160** and then add connections (e.g., relationships) to a number of other users of the social-networking system **160** whom they want to be connected to. Herein, the term "friend" may refer to any other user of the social-networking system **160** with whom a user has formed a connection, association, or relationship via the social-networking system **160**.
(22) In particular embodiments, the social-networking system **160** may provide users with the ability to take actions on various types of items or objects, supported by the social-networking system **160**. As an example and not by way of limitation, the items and objects may include groups or social networks to which users of the social-networking system **160** may belong, events or calendar entries in which a user might be interested, computer-based applications that a user may use, transactions that allow users to buy or sell items via the service, interactions with advertisements that a user may perform, or other suitable items or objects. A user may interact with anything that is capable of being represented in the social-networking system **160** or by an external system of a third-party system **170**, which is separate from the social-networking system **160** and coupled to the social-networking system **160** via a network **110**.
(23) In particular embodiments, the social-networking system **160** may be capable of linking a variety of entities. As an example and not by way of limitation, the social-networking system **160** may enable users to interact with each other as well as receive content from third-party systems **170** or other entities, or to allow users to interact with these entities through an application programming interfaces (API) or other communication channels.
(24) In particular embodiments, a third-party system **170** may include one or more types of servers, one or more data stores, one or more interfaces, including but not limited to APIs, one or more web services, one or more content sources, one or more networks, or any other suitable components, e.g., that servers may communicate with. A third-party system **170** may be operated by a different entity from an entity operating the social-networking system **160**. In particular embodiments, however, the social-networking system **160** and third-party systems **170** may operate in conjunction with each other to provide social-networking services to users of the social-networking system **160** or third-party systems **170**. In this sense, the social-networking system **160** may provide a platform, or backbone, which other systems, such as third-party systems **170**, may use to provide social-networking services and functionality to users across the Internet.
(25) In particular embodiments, a third-party system **170** may include a third-party content object provider. A third-party content object provider may include one or more sources of content objects, which may be communicated to a client system **130**. As an example and not by way of limitation, content objects may include information regarding things or activities of interest to the user, such as, for example, movie show times, movie reviews, restaurant reviews, restaurant menus, product

information and reviews, or other suitable information. As another example and not by way of limitation, content objects may include incentive content objects, such as coupons, discount tickets, gift certificates, or other suitable incentive objects.

(26) In particular embodiments, the social-networking system **160** also includes user-generated content objects, which may enhance a user's interactions with the social-networking system **160**. User-generated content may include anything a user can add, upload, send, or “post” to the social-networking system **160**. As an example and not by way of limitation, a user communicates posts to the social-networking system **160** from a client system **130**. Posts may include data such as status updates or other textual data, location information, photos, videos, links, music or other similar data or media. Content may also be added to the social-networking system **160** by a third-party through a “communication channel,” such as a newsfeed or stream.

(27) In particular embodiments, the social-networking system **160** may include a variety of servers, sub-systems, programs, modules, logs, and data stores. In particular embodiments, the social-networking system **160** may include one or more of the following: a web server, action logger, API-request server, relevance-and-ranking engine, content-object classifier, notification controller, action log, third-party-content-object-exposure log, inference module, authorization/privacy server, search module, advertisement-targeting module, user-interface module, user-profile store, connection store, third-party content store, or location store. The social-networking system **160** may also include suitable components such as network interfaces, security mechanisms, load balancers, failover servers, management-and-network-operations consoles, other suitable components, or any suitable combination thereof. In particular embodiments, the social-networking system **160** may include one or more user-profile stores for storing user profiles. A user profile may include, for example, biographic information, demographic information, behavioral information, social information, or other types of descriptive information, such as work experience, educational history, hobbies or preferences, interests, affinities, or location. Interest information may include interests related to one or more categories. Categories may be general or specific. As an example and not by way of limitation, if a user “likes” an article about a brand of shoes the category may be the brand, or the general category of “shoes” or “clothing.” A connection store may be used for storing connection information about users. The connection information may indicate users who have similar or common work experience, group memberships, hobbies, educational history, or are in any way related or share common attributes. The connection information may also include user-defined connections between different users and content (both internal and external). A web server may be used for linking the social-networking system **160** to one or more client systems **130** or one or more third-party systems **170** via a network **110**. The web server may include a mail server or other messaging functionality for receiving and routing messages between the social-networking system **160** and one or more client systems **130**. An API-request server may allow a third-party system **170** to access information from the social-networking system **160** by calling one or more APIs. An action logger may be used to receive communications from a web server about a user's actions on or off the social-networking system **160**. In conjunction with the action log, a third-party-content-object log may be maintained of user exposures to third-party-content objects. A notification controller may provide information regarding content objects to a client system **130**. Information may be pushed to a client system **130** as notifications, or information may be pulled from a client system **130** responsive to a request received from a client system **130**. Authorization servers may be used to enforce one or more privacy settings of the users of the social-networking system **160**. A privacy setting of a user determines how particular information associated with a user can be shared. The authorization server may allow users to opt in to or opt out of having their actions logged by the social-networking system **160** or shared with other systems (e.g., a third-party system **170**), such as, for example, by setting appropriate privacy settings. Third-party-content-object stores may be used to store content objects received from third parties, such as a third-party system **170**. Location stores may be used for storing location information received from

client systems **130** associated with users. Advertisement-pricing modules may combine social information, the current time, location information, or other suitable information to provide relevant advertisements, in the form of notifications, to a user.

(28) Assistant Systems

(29) FIG. **2** illustrates an example architecture of the assistant system **140**. In particular embodiments, the assistant system **140** may assist a user to obtain information or services. The assistant system **140** may enable the user to interact with it with multi-modal user input (such as voice, text, image, video) in stateful and multi-turn conversations to get assistance. The assistant system **140** may create and store a user profile comprising both personal and contextual information associated with the user. In particular embodiments, the assistant system **140** may analyze the user input using natural-language understanding. The analysis may be based on the user profile for more personalized and context-aware understanding. The assistant system **140** may resolve entities associated with the user input based on the analysis. In particular embodiments, the assistant system **140** may interact with different agents to obtain information or services that are associated with the resolved entities. The assistant system **140** may generate a response for the user regarding the information or services by using natural-language generation. Through the interaction with the user, the assistant system **140** may use dialog management techniques to manage and forward the conversation flow with the user. In particular embodiments, the assistant system **140** may further assist the user to effectively and efficiently digest the obtained information by summarizing the information. The assistant system **140** may also assist the user to be more engaging with an online social network by providing tools that help the user interact with the online social network (e.g., creating posts, comments, messages). The assistant system **140** may additionally assist the user to manage different tasks such as keeping track of events. In particular embodiments, the assistant system **140** may proactively execute pre-authorized tasks that are relevant to user interests and preferences based on the user profile, at a time relevant for the user, without a user input. In particular embodiments, the assistant system **140** may check privacy settings to ensure that accessing a user's profile or other user information and executing different tasks are permitted subject to the user's privacy settings.

(30) In particular embodiments, the assistant system **140** may receive a user input from the assistant application **136** in the client system **130** associated with the user. If the user input is based on a text modality, the assistant system **140** may receive it at a messaging platform **205**. If the user input is based on an audio modality (e.g., the user may speak to the assistant application **136** or send a video including speech to the assistant application **136**), the assistant system **140** may process it using an audio speech recognition (ASR) module **210** to convert the user input into text. If the user input is based on an image or video modality, the assistant system **140** may process it using optical character recognition techniques within the messaging platform **205** to convert the user input into text. The output of the messaging platform **205** or the ASR module **210** may be received at an assistant xbot **215**.

(31) In particular embodiments, the assistant xbot **215** may be a type of chat bot. The assistant xbot **215** may comprise a programmable service channel, which may be a software code, logic, or routine that functions as a personal assistant to the user. The assistant xbot **215** may work as the user's portal to the assistant system **140**. The assistant xbot **215** may therefore be considered as a type of conversational agent. In particular embodiments, the assistant xbot **215** may send the textual user input to a natural-language understanding (NLU) module **220** to interpret the user input. In particular embodiments, the NLU module **220** may get information from a user context engine **225** and a semantic information aggregator **230** to accurately understand the user input. The user context engine **225** may store the user profile of the user. The user profile of the user may comprise user-profile data including demographic information, social information, and contextual information associated with the user. The user-profile data may also include user interests and preferences on a plurality of topics, aggregated through conversations on news feed, search logs,

messaging platform **205**, etc. The usage of a user profile may be protected behind a privacy check module **245** to ensure that a user's information can be used only for his/her benefit, and not shared with anyone else. The semantic information aggregator **230** may provide ontology data associated with a plurality of predefined domains, intents, and slots to the NLU module **220**. In particular embodiments, a domain may denote a social context of interaction, e.g., education. An intent may indicate a purpose of a user interacting with the assistant system **140**. A slot may represent a basic semantic entity. For example, a slot for “pizza” may be dish. The semantic information aggregator **230** may additionally extract information from a social graph, a knowledge graph, and a concept graph, and retrieve a user's profile from the user context engine **225**. The semantic information aggregator **230** may further process information from these different sources by determining what information to aggregate, annotating n-grams of the user input, ranking the n-grams with confidence scores based on the aggregated information, formulating the ranked n-grams into features that can be used by the NLU module **220** for understanding the user input. Based on the output of the user context engine **225** and the semantic information aggregator **230**, the NLU module **220** may identify a domain, an intent, and one or more slots from the user input in a personalized and context-aware manner. In particular embodiments, the NLU module **220** may comprise a lexicon of language and a parser and grammar rules to partition sentences into an internal representation. The NLU module **220** may also comprise one or more programs that perform naive semantics or stochastic semantic analysis to the use of pragmatics to understand a user input. In particular embodiments, the parser may be based on a deep learning architecture comprising multiple long-short term memory (LSTM) networks. As an example and not by way of limitation, the parser may be based on a recurrent neural network grammar (RNNG) model, which is a type of recurrent and recursive LSTM algorithm.

(32) In particular embodiments, the identified domain, intent, and one or more slots from the NLU module **220** may be sent to a dialog engine **235**. In particular embodiments, the dialog engine **235** may manage the dialog state and flow of the conversation between the user and the assistant xbot **215**. The dialog engine **235** may additionally store previous conversations between the user and the assistant xbot **215**. In particular embodiments, the dialog engine **235** may communicate with an entity resolution module **240** to resolve entities associated with the one or more slots, which supports the dialog engine **235** to forward the flow of the conversation between the user and the assistant xbot **215**. In particular embodiments, the entity resolution module **240** may access the social graph, the knowledge graph, and the concept graph when resolving the entities. Entities may include, for example, unique users or concepts, each of which may have a unique identifier (ID). As an example and not by way of limitation, the knowledge graph may comprise a plurality of entities. Each entity may comprise a single record associated with one or more attribute values. The particular record may be associated with a unique entity identifier. Each record may have diverse values for an attribute of the entity. Each attribute value may be associated with a confidence probability. A confidence probability for an attribute value represents a probability that the value is accurate for the given attribute. Each attribute value may be also associated with a semantic weight. A semantic weight for an attribute value may represent how the value semantically appropriate for the given attribute considering all the available information. For example, the knowledge graph may comprise an entity of a movie “The Martian” (**2015**), which includes information that has been extracted from multiple content sources and then deduped, resolved, and fused to generate the single unique record for the knowledge graph. The entity may be associated with a space attribute value which indicates the genre of the movie “The Martian” (**2015**). The entity resolution module **240** may additionally request a user profile of the user associated with the user input from the user context engine **225**. In particular embodiments, the entity resolution module **240** may communicate with a privacy check module **245** to guarantee that the resolving of the entities does not violate privacy policies. In particular embodiments, the privacy check module **245** may use an authorization/privacy server to enforce privacy policies. As an example and not by way of

limitation, an entity to be resolved may be another user who specifies in his/her privacy settings that his/her identity should not be searchable on the online social network, and thus the entity resolution module **240** may not return that user's identifier in response to a request. Based on the information obtained from the social graph, knowledge graph, concept graph, and user profile, and subject to applicable privacy policies, the entity resolution module **240** may therefore accurately resolve the entities associated with the user input in a personalized and context-aware manner. In particular embodiments, each of the resolved entities may be associated with one or more identifiers hosted by the social-networking system **160**. As an example and not by way of limitation, an identifier may comprise a unique user identifier (ID). In particular embodiments, each of the resolved entities may be also associated with a confidence score.

(33) In particular embodiments, the dialog engine **235** may communicate with different agents based on the identified intent and domain, and the resolved entities. In particular embodiments, the agents may comprise first-party agents **250** and third-party agents **255**. In particular embodiments, first-party agents **250** may comprise internal agents that are accessible and controllable by the assistant system **140** (e.g. agents associated with services provided by the online social network). In particular embodiments, third-party agents **255** may comprise external agents that the assistant system **140** has no control over (e.g., music streams agents, ticket sales agents). The first-party agents **250** may be associated with first-party providers **260** that provide content objects and/or services hosted by the social-networking system **160**. The third-party agents **255** may be associated with third-party providers **265** that provide content objects and/or services hosted by the third-party system **170**.

(34) In particular embodiments, the communication from the dialog engine **235** to the first-party agents **250** may comprise requesting particular content objects and/or services provided by the first-party providers **260**. As a result, the first-party agents **250** may retrieve the requested content objects from the first-party providers **260** and/or execute tasks that command the first-party providers **260** to perform the requested services. In particular embodiments, the communication from the dialog engine **235** to the third-party agents **255** may comprise requesting particular content objects and/or services provided by the third-party providers **265**. As a result, the third-party agents **255** may retrieve the requested content objects from the third-party providers **265** and/or execute tasks that command the third-party providers **265** to perform the requested services. The third-party agents **255** may access the privacy check module **245** to guarantee no privacy violations before interacting with the third-party providers **265**. As an example and not by way of limitation, the user associated with the user input may specify in his/her privacy settings that his/her profile information is invisible to any third-party content providers. Therefore, when retrieving content objects associated with the user input from the third-party providers **265**, the third-party agents **255** may complete the retrieval without revealing to the third-party providers **265** which user is requesting the content objects.

(35) In particular embodiments, each of the first-party agents **250** or third-party agents **255** may be designated for a particular domain. As an example and not by way of limitation, the domain may comprise weather, transportation, music, etc. In particular embodiments, the assistant system **140** may use a plurality of agents collaboratively to respond to a user input. As an example and not by way of limitation, the user input may comprise "direct me to my next meeting." The assistant system **140** may use a calendar agent to retrieve the location of the next meeting. The assistant system **140** may then use a navigation agent to direct the user to the next meeting.

(36) In particular embodiments, each of the first-party agents **250** or third-party agents **255** may retrieve a user profile from the user context engine **225** to execute tasks in a personalized and context-aware manner. As an example and not by way of limitation, a user input may comprise "book me a ride to the airport." A transportation agent may execute the task of booking the ride. The transportation agent may retrieve the user profile of the user from the user context engine **225** before booking the ride. For example, the user profile may indicate that the user prefers taxis, so

the transportation agent may book a taxi for the user. As another example, the contextual information associated with the user profile may indicate that the user is in a hurry so the transportation agent may book a ride from a ride-sharing service for the user since it may be faster to get a car from a ride-sharing service than a taxi company. In particular embodiment, each of the first-party agents **250** or third-party agents **255** may take into account other factors when executing tasks. As an example and not by way of limitation, other factors may comprise price, rating, efficiency, partnerships with the online social network, etc.

(37) In particular embodiments, the dialog engine **235** may communicate with a conversational understanding composer (CU composer) **270**. The dialog engine **235** may send the requested content objects and/or the statuses of the requested services to the CU composer **270**. In particular embodiments, the dialog engine **235** may send the requested content objects and/or the statuses of the requested services as a <k, c, u, d> tuple, in which k indicates a knowledge source, c indicates a communicative goal, u indicates a user model, and d indicates a discourse model. In particular embodiments, the CU composer **270** may comprise a natural-language generator (NLG) **271** and a user interface (UI) payload generator **272**. The natural-language generator **271** may generate a communication content based on the output of the dialog engine **235**. In particular embodiments, the NLG **271** may comprise a content determination component, a sentence planner, and a surface realization component. The content determination component may determine the communication content based on the knowledge source, communicative goal, and the user's expectations. As an example and not by way of limitation, the determining may be based on a description logic. The description logic may comprise, for example, three fundamental notions which are individuals (representing objects in the domain), concepts (describing sets of individuals), and roles (representing binary relations between individuals or concepts). The description logic may be characterized by a set of constructors that allow the natural-language generator **271** to build complex concepts/roles from atomic ones. In particular embodiments, the content determination component may perform the following tasks to determine the communication content. The first task may comprise a translation task, in which the input to the natural-language generator **271** may be translated to concepts. The second task may comprise a selection task, in which relevant concepts may be selected among those resulted from the translation task based on the user model. The third task may comprise a verification task, in which the coherence of the selected concepts may be verified. The fourth task may comprise an instantiation task, in which the verified concepts may be instantiated as an executable file that can be processed by the natural-language generator **271**. The sentence planner may determine the organization of the communication content to make it human understandable. The surface realization component may determine specific words to use, the sequence of the sentences, and the style of the communication content. The UI payload generator **272** may determine a preferred modality of the communication content to be presented to the user. In particular embodiments, the CU composer **270** may communicate with the privacy check module **245** to make sure the generation of the communication content follows the privacy policies. In particular embodiments, the CU composer **270** may retrieve a user profile from the user context engine **225** when generating the communication content and determining the modality of the communication content. As a result, the communication content may be more natural, personalized, and context-aware for the user. As an example and not by way of limitation, the user profile may indicate that the user likes short sentences in conversations so the generated communication content may be based on short sentences. As another example and not by way of limitation, the contextual information associated with the user profile may indicated that the user is using a device that only outputs audio signals so the UI payload generator **272** may determine the modality of the communication content as audio.

(38) In particular embodiments, the CU composer **270** may send the generated communication content to the assistant xbot **215**. In particular embodiments, the assistant xbot **215** may send the communication content to the messaging platform **205**. The messaging platform **205** may further

send the communication content to the client system **130** via the assistant application **136**. In alternative embodiments, the assistant xbot **215** may send the communication content to a text-to-speech (TTS) module **275**. The TTS module **275** may convert the communication content to an audio clip. The TTS module **275** may further send the audio clip to the client system **130** via the assistant application **136**.

(39) In particular embodiments, the assistant xbot **215** may interact with a proactive inference layer **280** without receiving a user input. The proactive inference layer **280** may infer user interests and preferences based on the user profile that is retrieved from the user context engine **225**. In particular embodiments, the proactive inference layer **280** may further communicate with proactive agents **285** regarding the inference. The proactive agents **285** may execute proactive tasks based on the inference. As an example and not by way of limitation, the proactive tasks may comprise sending content objects or providing services to the user. In particular embodiments, each proactive task may be associated with an agenda item. The agenda item may comprise a recurring item such as a daily digest. The agenda item may also comprise a one-time item. In particular embodiments, a proactive agent **285** may retrieve the user profile from the user context engine **225** when executing the proactive task. Therefore, the proactive agent **285** may execute the proactive task in a personalized and context-aware manner. As an example and not by way of limitation, the proactive inference layer may infer that the user likes the band Maroon **5** and the proactive agent **285** may generate a recommendation of Maroon **5**'s new song/album to the user.

(40) In particular embodiments, the proactive agent **285** may generate candidate entities associated with the proactive task based on a user profile. The generation may be based on a straightforward backend query using deterministic filters to retrieve the candidate entities from a structured data store. The generation may be alternatively based on a machine-learning model that is trained based on the user profile, entity attributes, and relevance between users and entities. As an example and not by way of limitation, the machine-learning model may be based on support vector machines (SVM). As another example and not by way of limitation, the machine-learning model may be based on a regression model. As another example and not by way of limitation, the machine-learning model may be based on a deep convolutional neural network (DCNN). In particular embodiments, the proactive agent **285** may also rank the generated candidate entities based on the user profile and the content associated with the candidate entities. The ranking may be based on the similarities between a user's interests and the candidate entities. As an example and not by way of limitation, the assistant system **140** may generate a feature vector representing a user's interest and feature vectors representing the candidate entities. The assistant system **140** may then calculate similarity scores (e.g., based on cosine similarity) between the feature vector representing the user's interest and the feature vectors representing the candidate entities. The ranking may be alternatively based on a ranking model that is trained based on user feedback data.

(41) In particular embodiments, the proactive task may comprise recommending the candidate entities to a user. The proactive agent **285** may schedule the recommendation, thereby associating a recommendation time with the recommended candidate entities. The recommended candidate entities may be also associated with a priority and an expiration time. In particular embodiments, the recommended candidate entities may be sent to a proactive scheduler. The proactive scheduler may determine an actual time to send the recommended candidate entities to the user based on the priority associated with the task and other relevant factors (e.g., clicks and impressions of the recommended candidate entities). In particular embodiments, the proactive scheduler may then send the recommended candidate entities with the determined actual time to an asynchronous tier. The asynchronous tier may temporarily store the recommended candidate entities as a job. In particular embodiments, the asynchronous tier may send the job to the dialog engine **235** at the determined actual time for execution. In alternative embodiments, the asynchronous tier may execute the job by sending it to other surfaces (e.g., other notification services associated with the social-networking system **160**). In particular embodiments, the dialog engine **235** may identify the

dialog intent, state, and history associated with the user. Based on the dialog intent, the dialog engine **235** may select some candidate entities among the recommended candidate entities to send to the client system **130**. In particular embodiments, the dialog state and history may indicate if the user is engaged in an ongoing conversation with the assistant xbot **215**. If the user is engaged in an ongoing conversation and the priority of the task of recommendation is low, the dialog engine **235** may communicate with the proactive scheduler to reschedule a time to send the selected candidate entities to the client system **130**. If the user is engaged in an ongoing conversation and the priority of the task of recommendation is high, the dialog engine **235** may initiate a new dialog session with the user in which the selected candidate entities may be presented. As a result, the interruption of the ongoing conversation may be prevented. When it is determined that sending the selected candidate entities is not interruptive to the user, the dialog engine **235** may send the selected candidate entities to the CU composer **270** to generate a personalized and context-aware communication content comprising the selected candidate entities, subject to the user's privacy settings. In particular embodiments, the CU composer **270** may send the communication content to the assistant xbot **215** which may then send it to the client system **130** via the messaging platform **205** or the TTS module **275**.

(42) In particular embodiments, the assistant xbot **215** may communicate with a proactive agent **285** in response to a user input. As an example and not by way of limitation, the user may ask the assistant xbot **215** to set up a reminder. The assistant xbot **215** may request a proactive agent **285** to set up such reminder and the proactive agent **285** may proactively execute the task of reminding the user at a later time.

(43) In particular embodiments, the assistant system **140** may comprise a summarizer **290**. The summarizer **290** may provide customized news feed summaries to a user. In particular embodiments, the summarizer **290** may comprise a plurality of meta agents. The plurality of meta agents may use the first-party agents **250**, third-party agents **255**, or proactive agents **285** to generated news feed summaries. In particular embodiments, the summarizer **290** may retrieve user interests and preferences from the proactive inference layer **280**. The summarizer **290** may then retrieve entities associated with the user interests and preferences from the entity resolution module **240**. The summarizer **290** may further retrieve a user profile from the user context engine **225**. Based on the information from the proactive inference layer **280**, the entity resolution module **240**, and the user context engine **225**, the summarizer **290** may generate personalized and context-aware summaries for the user. In particular embodiments, the summarizer **290** may send the summaries to the CU composer **270**. The CU composer **270** may process the summaries and send the processing results to the assistant xbot **215**. The assistant xbot **215** may then send the processed summaries to the client system **130** via the messaging platform **205** or the TTS module **275**.

(44) FIG. **3** illustrates an example diagram flow of responding to a user request by the assistant system **140**. In particular embodiments, the assistant xbot **215** may access a request manager **305** upon receiving the user request. The request manager **305** may comprise a context extractor **306** and a conversational understanding object generator (CU object generator) **307**. The context extractor **306** may extract contextual information associated with the user request. The CU object generator **307** may generate particular content objects relevant to the user request. In particular embodiments, the request manager **305** may store the contextual information and the generated content objects in data store **310** which is a particular data store implemented in the assistant system **140**.

(45) In particular embodiments, the request manger **305** may send the generated content objects to the NLU module **220**. The NLU module **220** may perform a plurality of steps to process the content objects. At step **221**, the NLU module **220** may generate a whitelist for the content objects. At step **222**, the NLU module **220** may perform a featurization based on the whitelist. At step **223**, the NLU module **220** may perform domain classification/selection based on the features resulted from the featurization. The domain classification/selection results may be further processed based on two

related procedures. At step **224***a*, the NLU module **220** may process the domain classification/selection result using an intent classifier. The intent classifier may determine the user's intent associated with the user request. As an example and not by way of limitation, the intent classifier may be based on a machine-learning model that may take the domain classification/selection result as input and calculate a probability of the input being associated with a particular predefined intent. At step **224***b*, the NLU module may process the domain classification/selection result using a meta-intent classifier. The meta-intent classifier may determine categories that describe the user's intent. As an example and not by way of limitation, the meta-intent classifier may be based on a machine-learning model that may take the domain classification/selection result as input and calculate a probability of the input being associated with a particular predefined meta-intent. At step **225***a*, the NLU module **220** may use a slot tagger to annotate one or more slots associated with the user request. At step **225***b*, the NLU module **220** may use a meta slot tagger to annotate one or more slots for the classification result from the meta-intent classifier. As an example and not by way of limitation, a user request may comprise "change 500 dollars in my account to Japanese yen." The intent classifier may take the user request as input and formulate it into a vector. The intent classifier may then calculate probabilities of the user request being associated with different predefined intents based on a vector comparison between the vector representing the user request and the vectors representing different predefined intents. In a similar manner, the slot tagger may take the user request as input and formulate each word into a vector. The intent classifier may then calculate probabilities of each word being associated with different predefined slots based on a vector comparison between the vector representing the word and the vectors representing different predefined slots. The intent of the user may be classified as "changing money". The slots of the user request may comprise "500", "dollars", "account", and "Japanese yen". The meta-intent of the user may be classified as "financial service". The meta slot may comprise "finance".

(46) In particular embodiments, the NLU module **220** may improve the domain classification/selection of the content objects by extracting semantic information from the semantic information aggregator **230**. In particular embodiments, the semantic information aggregator **230** may aggregate semantic information in the following way. The semantic information aggregator **230** may first retrieve information from the user context engine **225**. In particular embodiments, the user context engine **225** may comprise offline aggregators **226** and an online inference service **227**. The offline aggregators **226** may process a plurality of data associated with the user that are collected from a prior time window. As an example and not by way of limitation, the data may include news feed posts/comments, interactions with news feed posts/comments, search history, etc. that are collected from a prior 90-day window. The processing result may be stored in the user context engine **225** as part of the user profile. The online inference service **227** may analyze the conversational data associated with the user that are received by the assistant system **140** at a current time. The analysis result may be stored in the user context engine **225** also as part of the user profile. In particular embodiments, the semantic information aggregator **230** may then process the retrieved information, i.e., a user profile, from the user context engine **225** in the following steps. At step **231**, the semantic information aggregator **230** may process the retrieved information from the user context engine **225** based on natural-language processing (NLP). At step **232**, the processing result may be annotated with entities by an entity tagger. Based on the annotations, the semantic information aggregator **230** may generate dictionaries for the retrieved information at step **233**. At step **234**, the semantic information aggregator **230** may rank the entities tagged by the entity tagger. In particular embodiments, the semantic information aggregator **230** may communicate with different graphs **330** including social graph, knowledge graph, and concept graph to extract ontology data that is relevant to the retrieved information from the user context engine **225**. In particular embodiments, the semantic information aggregator **230** may aggregate the user profile, the ranked entities, and the information from the graphs **330**. The semantic

information aggregator **230** may then send the aggregated information to the NLU module **220** to facilitate the domain classification/selection.

(47) In particular embodiments, the output of the NLU module **220** may be sent to a co-reference module **315** to interpret references of the content objects associated with the user request. The co-reference module **315** may comprise reference creation **316** and reference resolution **317**. In particular embodiments, the reference creation **316** may create references for entities determined by the NLU module **220**. The reference resolution **317** may resolve these references accurately. In particular embodiments, the co-reference module **315** may access the user context engine **225** and the dialog engine **235** when necessary to interpret references with improved accuracy.

(48) In particular embodiments, the identified domains, intents, meta-intents, slots, and meta slots, along with the resolved references may be sent to the entity resolution module **240** to resolve relevant entities. In particular embodiments, the entity resolution module **240** may comprise domain entity resolution **241** and generic entity resolution **242**. The domain entity resolution **241** may resolve the entities by categorizing the slots and meta slots into different domains. In particular embodiments, entities may be resolved based on the ontology data extracted from the graphs **330**. The ontology data may comprise the structural relationship between different slots/meta-slots and domains. The ontology may also comprise information of how the slots/meta-slots may be grouped, related within a hierarchy where the higher level comprises the domain, and subdivided according to similarities and differences. The generic entity resolution **242** may resolve the entities by categorizing the slots and meta slots into different generic topics. In particular embodiments, the resolving may be also based on the ontology data extracted from the graphs **330**. The ontology data may comprise the structural relationship between different slots/meta-slots and generic topics. The ontology may also comprise information of how the slots/meta-slots may be grouped, related within a hierarchy where the higher level comprises the topic, and subdivided according to similarities and differences. As an example and not by way of limitation, in response to the input of an inquiry of the advantages of a car, the generic entity resolution **242** may resolve the car as vehicle and the domain entity resolution **241** may resolve the car as electric car.

(49) In particular embodiments, the output of the entity resolution module **240** may be sent to the dialog engine **235** to forward the flow of the conversation with the user. The dialog engine **235** may comprise dialog intent resolution **236** and dialog state update/ranker **237**. In particular embodiments, the dialog intent resolution **236** may resolve the user intent associated with the current dialog session. In particular embodiments, the dialog state update/ranker **237** may update/rank the state of the current dialog session. As an example and not by way of limitation, the dialog state update/ranker **237** may update the dialog state as “completed” if the dialog session is over. As another example and not by way of limitation, the dialog state update/ranker **237** may rank the dialog state based on a priority associated with it.

(50) In particular embodiments, the dialog engine **235** may communicate with a task completion module **335** about the dialog intent and associated content objects. The task completion module **335** may comprise an action selection component **336**. In particular embodiments, the dialog engine **235** may additionally check against dialog policies **320** regarding the dialog state. The dialog policies **320** may comprise generic policy **321** and domain specific policies **322**, both of which may guide how to select the next system action based on the dialog state. In particular embodiments, the task completion module **335** may communicate with dialog policies **320** to obtain the guidance of the next system action. In particular embodiments, the action selection component **336** may therefore select an action based on the dialog intent, the associated content objects, and the guidance from dialog policies **320**.

(51) In particular embodiments, the output of the task completion module **335** may be sent to the CU composer **270**. In alternative embodiments, the selected action may require one or more agents **340** to be involved. As a result, the task completion module **335** may inform the agents **340** about the selected action. Meanwhile, the dialog engine **235** may receive an instruction to update the

dialog state. As an example and not by way of limitation, the update may comprise awaiting agents' response. In particular embodiments, the CU composer **270** may generate a communication content for the user using the NLG **271** based on the output of the task completion module **335**. The CU composer **270** may also determine a modality of the generated communication content using the UI payload generator **272**. Since the generated communication content may be considered as a response to the user request, the CU composer **270** may additionally rank the generated communication content using a response ranker **273**. As an example and not by way of limitation, the ranking may indicate the priority of the response.

(52) In particular embodiments, the output of the CU composer **270** may be sent to a response manager **325**. The response manager **325** may perform different tasks including storing/updating the dialog state **326** retrieved from data store **310** and generating responses **327**. In particular embodiments, the generated response and the communication content may be sent to the assistant xbot **215**. In alternative embodiments, the output of the CU composer **270** may be additionally sent to the TTS module **275** if the determined modality of the communication content is audio. The speech generated by the TTS module **275** and the response generated by the response manager **325** may be then sent to the assistant xbot **215**.

(53) Processing Multimodal User Input

(54) In particular embodiments, the assistant system **140** may receive a multimodal user input (e.g., sound, image, video, text) from a client system **130** associated with a user. The assistant system **140** may process the multimodal user input with different modules of the assistant system **140** based on the different modalities of the user input. The assistant system **140** may then understand the user input, i.e., identify subjects/entities and semantic meaning of the user input, based on an entity resolution module **240** and a co-reference module **315**. The assistant system **140** may further generate a multimodal output based on the understanding and send the multimodal output to the user via the client system **130**. As an example and not by way of limitation, a user may submit a picture of several people to the assistant system **140** and ask the assistant system **140** to initiate a call to one of them (e.g., the user may say “call the guy on the left”). The assistant system **140** may determine the identities of these people, identify the person “on the left,” access the contact information of the person on the left (subject to that person's privacy settings), and initiate a call for the user accordingly. In particular, the assistant system **140** may use a visual-recognition agent capable of processing visual input included in the multimodal user input, where the visual-recognition agent may access various computer-vision models to understand the visual input. The understanding of the visual input may be used by the co-reference module **315** to resolve the entities associated with the user input. Based on the understanding of the visual input and the resolved entities, the assistant system **140** may use an assistant xbot **215** to conduct a conversation with the user to further enhance the user's experience with the assistant system **140** by executing different tasks corresponding to the visual input. The assistant system **140** may reactively process multimodal user input in response to a user query. The assistant system **140** may also proactively process a multimodal user input by making suggestions based on the user input. The capability of processing multimodal user input and seamlessly switching between different modalities of output makes the assistant system **140** well adapted to various client systems **130** and software, thus making it more useful and appealing to users. Although this disclosure describes processing particular multimodal user input via a particular system in a particular manner, this disclosure contemplates processing any suitable multimodal user input via any suitable system in any suitable manner.

(55) In particular embodiments, the assistant system **140** may receive, from a client system **130** associated with a first user, a user input based on one or more modalities. At least one of the modalities of the user input may be a visual modality. In particular embodiments, the assistant system **140** may identify, based on one or more machine-learning models, one or more subjects associated with the user input based on the visual modality. In particular embodiments, the assistant

system **140** may determine, based on the one or more machine-learning models, one or more attributes associated with the one or more subjects, respectively. The assistant system **140** may then resolve, based on the determined one or more attributes, one or more entities corresponding to the one or more subjects. In particular embodiments, the assistant system **140** may execute one or more tasks associated with the one or more resolved entities. The assistant system **140** may further send, to the client system **130** associated with the first user, instructions for presenting a communication content responsive to user input. The communication content may comprise information associated with the executed one or more tasks.

(56) FIG. **4** illustrates an example diagram flow of processing multimodal user input based on the example architecture **200** of the assistant system **140** in FIG. **2**. In particular embodiments, the messaging platform **205** may receive the multimodal user input **405** from the client system **130**. The multimodal user input **405** may comprise one or more of a character string, an audio clip, an image, or a video clip. In particular embodiments, the assistant xbot **215** may receive the multimodal user input **405** from the messaging platform **205**. The assistant xbot **215** may identify that the multimodal user input **405** comprises a visual input **410** and a textual input **415**. In particular embodiments, the assistant xbot **215** may send the visual input **410** to one or more visual-recognition agents **450** to perform visual analysis of the visual input **410** to identify one or more subjects. In particular embodiments, a visual-recognition agent **450** may be a first-party agent **250**. In alternative embodiments, a visual-recognition agent **450** may be a third-party agent **255**. As an example and not by way of limitation, the one or more subjects associated with the user input **405** may comprise one or more of a person, a location, a business, or an object. In particular embodiments, the visual-recognition agents **450** may identify the one or more people based on facial recognition. In particular embodiments, the visual-recognition agents **450** may identify the one or more objects based on object detection. In particular embodiments, the visual-recognition agents **450** may store the identified one or more subjects in a dialog state in the dialog engine **235**. As a result, the assistant system **140** may efficiently retrieve the identified one or more subjects from the dialog engine **235** in the future when necessary. In particular embodiments, the visual-recognition agents **450** may additionally determine one or more attributes associated with the one or more subjects, respectively. As an example and not by way of limitation, the one or more attributes may be a position within an image (e.g., on the left), a color associated with the subject, a shape of the subject, etc. In particular embodiments, the visual-recognition agents **450** may access one or more computer vision libraries for the visual analysis. As an example and not by way of limitation, the one or more computer vision libraries may comprise a library which is used to detect objects in images. More information on object detection may be found in U.S. patent application Ser. No. 15/277,938, filed 27 Sep. 2016, which is incorporated by reference. As another example and not by way of limitation, the one or more computer vision libraries may comprise a library which is used to recognize faces in images. More information on facial recognition may be found in U.S. patent application Ser. No. 12/922,984, filed 4 Mar. 2014, which is incorporated by reference. Using machine-learning models based on facial recognition and object detection to recognize people, locations, businesses, and objects together with their attributes may be an effective solution to address the technical challenge of accurately identifying, from visual input, subjects and their attributes. In particular embodiments, the visual-recognition agents **450** may generate a feature representation for the user input **405** based on the visual modality, i.e., the visual input **410**. The generated feature representation may be processed by one or more machine-learning models provided by the one or more computer vision libraries. As an example and not by way of limitation, the one or more machine-learning models may comprise one or more of a support vector machine, a regression model, or a convolutional neural network. In particular embodiments, the visual-recognition agents **450** may communicate with the privacy check module **245** to make sure the visual analysis is subject to users' privacy settings when accessing the computer vision libraries. In particular embodiments, each library may have its own privacy check component, which

guarantees that object detection or facial recognition is subject to users' privacy settings. Although this disclosure describes particular visual analysis in particular manners, this disclosure contemplates any suitable visual analysis in any suitable manner.

(57) In particular embodiments, the assistant xbot **215** may send the textual input **415** to the NLU module **220**. The NLU module **220** may identify one or more intents and one or more slots **415** based on the textual input **415** of the user input **405**. In particular embodiments, the identified one or more intents and slots **420** and the visual analysis result **425** from the visual-recognition agents **450** may be sent to the co-reference module **315**. The co-reference module **315** may resolve one or more entities corresponding to the one or more subjects identified by the visual-recognition agents **450** based on the identified one or more slots. The co-reference module **315** is able to link the wording information of the textual input **415** or audio input with the visual analysis result **425** of the visual input **410**, which makes it an effective solution for addressing the technical challenge of accurately resolving entities from multimodal user input **405**. On the other hand, the joint analysis of the textual input **415** and visual input **410** via the co-reference module **315** may result in a technical advantage of improved understanding of multimodal user input **405**. In particular embodiments, the resolved entities and intents **430** may be sent to another one or more first-party agents **250** or third-party agents **255** which may execute one or more tasks associated with the one or more resolved entities based on the identified intents. As an example and not by way of limitation, a user input **405** may comprise an image and a sentence which is “call the person on the left”. The NLU module **220** may identify an intent as [IN:call(person)] and a person slot [SL:person “on the left”]. The visual-recognition agents **450** may perform visual analysis of the image to identify the person on the left in the image. Then the co-reference module **315** may resolve the identity of the person on the left. The assistant system **140** may further execute a task of calling the person accordingly. Although this disclosure describes resolving particular entities in particular manners, this disclosure contemplates resolving any suitable entities in any suitable manner.

(58) In particular embodiments, the first-party agents **250** or third-party agents **255** may send the status of the one or more tasks **435** to the CU composer **270**. The CU composer **270** may then generate a communication content **440** of information associated with the executed one or more tasks. The communication content **440** may comprise one or more of a character string, an audio clip, an image, or a video clip. In particular embodiments, the assistant system **140** may determine one or more modalities for the communication content **440** in the following way. The assistant system **140** may first identify contextual information associated with the first user. The assistant system **140** may then identify contextual information associated with the client system **130**. The assistant system **140** may further determine the one or more modalities based on the contextual information associated with the first user and the contextual information associated with the client system **130**. As an example and not by way of limitation, the contextual information associated with the user may indicate that the user is at home, which means any modalities may be suitable for the communication content **440**. On the other hand, the contextual information associated with the client system **130** may indicate that the client system **130** only supports audio output. As a result, the determined modalities may only comprise audio. As another example and not by way of limitation, the contextual information associated with the user may indicate that the user is traveling in a high-speed train and the contextual information associated with the client system **130** may indicate that the client system **130** is not connected to Wi-Fi but only cellular data. Under such condition, even though the client system **130** supports all the modalities, the assistant system **140** may determine text as the modality because downloading images and/or videos takes much longer time than text using cellular data. Determining proper modalities based on contextual information associated with both a user and a client system **130** associated with the user may be an effective solution for addressing the technical challenge of responding to multimodal user input **405** with communication contents **440** that are based on suitable modalities, which allows the assistant

system **140** to communicate with the user based on the determined modalities that may be more suitable for the user's current situation. On the other hand, the ability of processing multimodal user input **405** and switching between different modalities of output may result in a technical advantage of efficiently adapting to various client systems **130** and software. Although this disclosure describes determining particular modalities in particular manners, this disclosure contemplates determining any suitable modalities in any suitable manner.

(59) In particular embodiments, the user input **405** may comprise a user interaction with a media content object. The use interaction may vary based on which surface the assistant system **140** is integrated with. As an example and not by way of limitation, the assistant system **140** may be integrated with a photo library stored in the client system **130** or associated with an online social network. A user interaction, correspondingly, may be long-pressing a photo, which may trigger the assistant system **140** to perform the understanding of the pressed photo. As another example and not by way of limitation, the assistant system **140** may be integrated with a messaging application. A user interaction, correspondingly, may be sharing a photo in a message thread, which may trigger the assistant system **140** to perform the understanding of the shared photo. As another example and not by way of limitation, the assistant system **140** may be integrated with a live streaming surface. During the live streaming, the user may ask the assistant system **140** questions, which may act as a user input **405** to trigger the understanding of the live streaming data. For example, the user may be live streaming his/her visit of Louvre and ask the assistant system **140** which artist created the painting that is currently being live streamed. As another example and not by way of limitation, the assistant system **140** may be integrated with a virtual reality (VR) surface. For example, a user may be using a VR surface to explore New York City. The user may see a purse and ask the assistant system **140** about the brand and price of the purse, which may act as a user input **405**. As another example and not by way of limitation, the assistant system **140** may be integrated with a social-networking application. The user may be browsing posts of his/her friends and ask the assistant system **140** about a particular post of food, which may act as a user input **405** to trigger the assistant system **140** to get information for the user regarding which restaurant serves the food. Although this disclosure describes particular user interactions with particular media content objects in particular manners, this disclosure contemplates any suitable user interactions with any suitable media content objects in any suitable manner.

(60) In particular embodiments, the assistant system **140** may proactively generate a plurality of tasks based on the visual modality of the user input **405**. The assistant system **140** may then receive, from the client system **130** associated with the first user, a user selection of the one or more tasks from the plurality of tasks by the first user. As an example and not by way of limitation, a user may be browsing pictures of a new Nike shoe. The assistant system **140** may proactively generate a plurality of tasks that the user can select to view the price, purchase it, or view similar shoes. As another example and not by way of limitation, a user may be live streaming at home and the assistant system **140** may detect smoke in the background. The assistant system **140** may then proactively alert the user about the smoke to ensure safety. The assistant system **140** may further generate a few tasks including calling an emergency number, turning off electricity at home, etc. As another example and not by way of limitation, a client system **130** may have a camera that captures the real-time visual data within a user's home. The user's kid may walk in and, based on the detection of the kid, the assistant system **140** may proactively generate a few tasks for the kid. The tasks may include playing a game based on the kid's previous gaming history. As another example and not by way of limitation, a user may have shared a screenshot of a booking of a flight in a group messaging surface to his/her friends who will be traveling with him/her. The assistant system **140** may identify the flight information and proactively generate a few tasks including setting up a reminder for the flight, booking a hotel room of the destination, checking the weather forecast of the destination, renting a car of the destination, etc. As another example and not by way of limitation, a user may have shared a trailer of a new movie to his/her friends and the assistant

system **140** may proactively generate a few tasks for the user to select. The generated tasks may include buying tickets for the movie, posting the trailer to the user's news feed, and getting more information of the movie such as cast and rating. Proactively suggesting tasks to users based on analysis of the visual input **410** may result in a technical advantage of increasing the degree of users engaging with the assistant system **140**. Although this disclosure describes proactively generating particular tasks based on particular user input in particular manners, this disclosure contemplates generating any suitable tasks based on any suitable user input in any suitable manner.

(61) FIG. **5** illustrates an example interaction between a user and the assistant system **140** via multimodal user input **405** and system output. As illustrated in FIG. **5**, a user may submit a visual input **410**, which is a photo of three people. The assistant system **140** may perform a visual analysis of the photo to generate a visual analysis result **425** identifying the people in the photo, i.e., "People in photo: David Jones, Mark Smith, and Bill Anders". The user may also submit a textual input **415***a*, i.e., "Where does Mark live". The assistant system **140** may execute a task of determining Mark's residence based on the textual input **415***a* and the visual analysis result **425** (subject to Mark's privacy settings). The assistant system **140** may further generate a communication content **440***a* accordingly and present it to the user, i.e., "Your friend currently lives in Hometown U.S.A.". The user may submit another textual input **415***b* inquiring about Mark's recent activities, i.e., by inputting "What's new about Mark". Subsequently, the assistant system **140** may execute a task of summarizing Mark's recent activities based on the textual input **415***b* and the visual analysis result **425**. The assistant system **140** may further generate a communication content **440***b* accordingly and present it to the user, i.e., by presenting a horizontal scrolling list of summaries of posts on the online social network in which Mark Smith is referenced. The user may continue with another textual input **415***c* trying to call a person in the photo, i.e., by inputting "Call him". In this case, the assistant system **140** may generate a communication content **440***c* to interact with user to disambiguate which person the user wants to call, i.e., by responding "Who do you want to call?". The user may subsequently make a clarification with a textual input **415***d*, i.e., by inputting "David". The assistant system **140** may execute the task of calling the clarified person and generate a communication content **440***d* notifying the user of the status of the task **435**, i.e., by responding "Calling". The user may submit another user input **415***e* trying to set up a reminder, i.e., by inputting "Remind me to call Mark on his birthday". In a similar manner, the assistant system **140** may generate a communication content **440***e* to interact with user to get more details about the reminder from the user, i.e., by responding "What time do you want to set the reminder for?". The user may then provide such details with a textual input **415***f*, i.e., "Midnight". To this end, the assistant system **140** may executing the task of setting up the reminder and generate a communication content **440***f* notifying the user of the status of the task **435**, i.e., by responding "I have set a reminder for midnight, September 13 to call Mark on his birthday". Although this disclosure describes particular interactions between a user and a particular system in particular manners, this disclosure contemplates any suitable interactions between a user and any suitable system in any suitable manner.

(62) FIG. **6** illustrates an example method **600** for processing multimodal user input **405**. The method may begin at step **610**, where the assistant system **140** may receive, from a client system **130** associated with a first user, a user input **405** based on one or more modalities, wherein at least one of the modalities of the user input **405** is a visual modality. At step **620**, the assistant system **140** may identify, based on one or more machine-learning models, one or more subjects associated with the user input **405** based on the visual modality. At step **630**, the assistant system **140** may determine, based on the one or more machine-learning models, one or more attributes associated with the one or more subjects, respectively. At step **640**, the assistant system **140** may resolve, based on the determined one or more attributes, one or more entities corresponding to the one or more subjects. At step **650**, the assistant system **140** may execute one or more tasks associated with the one or more resolved entities. At step **660**, the assistant system **140** may send, to the client

system **130** associated with the first user, instructions for presenting a communication content **440** responsive to the user input **405**, wherein the communication content **440** comprises information associated with the executed one or more tasks. Particular embodiments may repeat one or more steps of the method of FIG. **6**, where appropriate. Although this disclosure describes and illustrates particular steps of the method of FIG. **6** as occurring in a particular order, this disclosure contemplates any suitable steps of the method of FIG. **6** occurring in any suitable order. Moreover, although this disclosure describes and illustrates an example method for processing multimodal user input, including the particular steps of the method of FIG. **6**, this disclosure contemplates any suitable method for processing multimodal user input, including any suitable steps, which may include all, some, or none of the steps of the method of FIG. **6**, where appropriate. Furthermore, although this disclosure describes and illustrates particular components, devices, or systems carrying out particular steps of the method of FIG. **6**, this disclosure contemplates any suitable combination of any suitable components, devices, or systems carrying out any suitable steps of the method of FIG. **6**.
(63) Social Graphs
(64) FIG. **7** illustrates an example social graph **700**. In particular embodiments, the social-networking system **160** may store one or more social graphs **700** in one or more data stores. In particular embodiments, the social graph **700** may include multiple nodes-which may include multiple user nodes **702** or multiple concept nodes **704**—and multiple edges **706** connecting the nodes. Each node may be associated with a unique entity (i.e., user or concept), each of which may have a unique identifier (ID), such as a unique number or username. The example social graph **700** illustrated in FIG. **7** is shown, for didactic purposes, in a two-dimensional visual map representation. In particular embodiments, a social-networking system **160**, a client system **130**, an assistant system **140**, or a third-party system **170** may access the social graph **700** and related social-graph information for suitable applications. The nodes and edges of the social graph **700** may be stored as data objects, for example, in a data store (such as a social-graph database). Such a data store may include one or more searchable or queryable indexes of nodes or edges of the social graph **700**.
(65) In particular embodiments, a user node **702** may correspond to a user of the social-networking system **160** or the assistant system **140**. As an example and not by way of limitation, a user may be an individual (human user), an entity (e.g., an enterprise, business, or third-party application), or a group (e.g., of individuals or entities) that interacts or communicates with or over the social-networking system **160** or the assistant system **140**. In particular embodiments, when a user registers for an account with the social-networking system **160**, the social-networking system **160** may create a user node **702** corresponding to the user, and store the user node **702** in one or more data stores. Users and user nodes **702** described herein may, where appropriate, refer to registered users and user nodes **702** associated with registered users. In addition or as an alternative, users and user nodes **702** described herein may, where appropriate, refer to users that have not registered with the social-networking system **160**. In particular embodiments, a user node **702** may be associated with information provided by a user or information gathered by various systems, including the social-networking system **160**. As an example and not by way of limitation, a user may provide his or her name, profile picture, contact information, birth date, sex, marital status, family status, employment, education background, preferences, interests, or other demographic information. In particular embodiments, a user node **702** may be associated with one or more data objects corresponding to information associated with a user. In particular embodiments, a user node **702** may correspond to one or more web interfaces.
(66) In particular embodiments, a concept node **704** may correspond to a concept. As an example and not by way of limitation, a concept may correspond to a place (such as, for example, a movie theater, restaurant, landmark, or city); a website (such as, for example, a website associated with the social-networking system **160** or a third-party website associated with a web-application

server); an entity (such as, for example, a person, business, group, sports team, or celebrity); a resource (such as, for example, an audio file, video file, digital photo, text file, structured document, or application) which may be located within the social-networking system **160** or on an external server, such as a web-application server; real or intellectual property (such as, for example, a sculpture, painting, movie, game, song, idea, photograph, or written work); a game; an activity; an idea or theory; another suitable concept; or two or more such concepts. A concept node **704** may be associated with information of a concept provided by a user or information gathered by various systems, including the social-networking system **160** and the assistant system **140**. As an example and not by way of limitation, information of a concept may include a name or a title; one or more images (e.g., an image of the cover page of a book); a location (e.g., an address or a geographical location); a website (which may be associated with a URL); contact information (e.g., a phone number or an email address); other suitable concept information; or any suitable combination of such information. In particular embodiments, a concept node **704** may be associated with one or more data objects corresponding to information associated with concept node **704**. In particular embodiments, a concept node **704** may correspond to one or more web interfaces.

(67) In particular embodiments, a node in the social graph **700** may represent or be represented by a web interface (which may be referred to as a “profile interface”). Profile interfaces may be hosted by or accessible to the social-networking system **160** or the assistant system **140**. Profile interfaces may also be hosted on third-party websites associated with a third-party system **170**. As an example and not by way of limitation, a profile interface corresponding to a particular external web interface may be the particular external web interface and the profile interface may correspond to a particular concept node **704**. Profile interfaces may be viewable by all or a selected subset of other users. As an example and not by way of limitation, a user node **702** may have a corresponding user-profile interface in which the corresponding user may add content, make declarations, or otherwise express himself or herself. As another example and not by way of limitation, a concept node **704** may have a corresponding concept-profile interface in which one or more users may add content, make declarations, or express themselves, particularly in relation to the concept corresponding to concept node **704**.

(68) In particular embodiments, a concept node **704** may represent a third-party web interface or resource hosted by a third-party system **170**. The third-party web interface or resource may include, among other elements, content, a selectable or other icon, or other inter-actable object (which may be implemented, for example, in JavaScript, AJAX, or PHP codes) representing an action or activity. As an example and not by way of limitation, a third-party web interface may include a selectable icon such as “like,” “check-in,” “eat,” “recommend,” or another suitable action or activity. A user viewing the third-party web interface may perform an action by selecting one of the icons (e.g., “check-in”), causing a client system **130** to send to the social-networking system **160** a message indicating the user's action. In response to the message, the social-networking system **160** may create an edge (e.g., a check-in-type edge) between a user node **702** corresponding to the user and a concept node **704** corresponding to the third-party web interface or resource and store edge **706** in one or more data stores.

(69) In particular embodiments, a pair of nodes in the social graph **700** may be connected to each other by one or more edges **706**. An edge **706** connecting a pair of nodes may represent a relationship between the pair of nodes. In particular embodiments, an edge **706** may include or represent one or more data objects or attributes corresponding to the relationship between a pair of nodes. As an example and not by way of limitation, a first user may indicate that a second user is a “friend” of the first user. In response to this indication, the social-networking system **160** may send a “friend request” to the second user. If the second user confirms the “friend request,” the social-networking system **160** may create an edge **706** connecting the first user's user node **702** to the second user's user node **702** in the social graph **700** and store edge **706** as social-graph information in one or more of data stores **167**. In the example of FIG. **7**, the social graph **700** includes an edge

706 indicating a friend relation between user nodes **702** of user "A" and user "B" and an edge indicating a friend relation between user nodes **702** of user "C" and user "B." Although this disclosure describes or illustrates particular edges **706** with particular attributes connecting particular user nodes **702**, this disclosure contemplates any suitable edges **706** with any suitable attributes connecting user nodes **702**. As an example and not by way of limitation, an edge **706** may represent a friendship, family relationship, business or employment relationship, fan relationship (including, e.g., liking, etc.), follower relationship, visitor relationship (including, e.g., accessing, viewing, checking-in, sharing, etc.), subscriber relationship, superior/subordinate relationship, reciprocal relationship, non-reciprocal relationship, another suitable type of relationship, or two or more such relationships. Moreover, although this disclosure generally describes nodes as being connected, this disclosure also describes users or concepts as being connected. Herein, references to users or concepts being connected may, where appropriate, refer to the nodes corresponding to those users or concepts being connected in the social graph **700** by one or more edges **706**.

(70) In particular embodiments, an edge **706** between a user node **702** and a concept node **704** may represent a particular action or activity performed by a user associated with user node **702** toward a concept associated with a concept node **704**. As an example and not by way of limitation, as illustrated in FIG. **7**, a user may "like," "attended," "played," "listened," "cooked," "worked at," or "watched" a concept, each of which may correspond to an edge type or subtype. A concept-profile interface corresponding to a concept node **704** may include, for example, a selectable "check in" icon (such as, for example, a clickable "check in" icon) or a selectable "add to favorites" icon. Similarly, after a user clicks these icons, the social-networking system **160** may create a "favorite" edge or a "check in" edge in response to a user's action corresponding to a respective action. As another example and not by way of limitation, a user (user "C") may listen to a particular song ("SongName") using a particular application (an online music application). In this case, the social-networking system **160** may create a "listened" edge **706** and a "used" edge (as illustrated in FIG. **7**) between user nodes **702** corresponding to the user and concept nodes **704** corresponding to the song and application to indicate that the user listened to the song and used the application. Moreover, the social-networking system **160** may create a "played" edge **706** (as illustrated in FIG. **7**) between concept nodes **704** corresponding to the song and the application to indicate that the particular song was played by the particular application. In this case, "played" edge **706** corresponds to an action performed by an external application on an external audio file (the song "SongName"). Although this disclosure describes particular edges **706** with particular attributes connecting user nodes **702** and concept nodes **704**, this disclosure contemplates any suitable edges **706** with any suitable attributes connecting user nodes **702** and concept nodes **704**. Moreover, although this disclosure describes edges between a user node **702** and a concept node **704** representing a single relationship, this disclosure contemplates edges between a user node **702** and a concept node **704** representing one or more relationships. As an example and not by way of limitation, an edge **706** may represent both that a user likes and has used at a particular concept. Alternatively, another edge **706** may represent each type of relationship (or multiples of a single relationship) between a user node **702** and a concept node **704** (as illustrated in FIG. **7** between user node **702** for user "E" and concept node **704**).

(71) In particular embodiments, the social-networking system **160** may create an edge **706** between a user node **702** and a concept node **704** in the social graph **700**. As an example and not by way of limitation, a user viewing a concept-profile interface (such as, for example, by using a web browser or a special-purpose application hosted by the user's client system **130**) may indicate that he or she likes the concept represented by the concept node **704** by clicking or selecting a "Like" icon, which may cause the user's client system **130** to send to the social-networking system **160** a message indicating the user's liking of the concept associated with the concept-profile interface. In response to the message, the social-networking system **160** may create an edge **706** between user node **702** associated with the user and concept node **704**, as illustrated by "like" edge **706** between the user

and concept node **704**. In particular embodiments, the social-networking system **160** may store an edge **706** in one or more data stores. In particular embodiments, an edge **706** may be automatically formed by the social-networking system **160** in response to a particular user action. As an example and not by way of limitation, if a first user uploads a picture, watches a movie, or listens to a song, an edge **706** may be formed between user node **702** corresponding to the first user and concept nodes **704** corresponding to those concepts. Although this disclosure describes forming particular edges **706** in particular manners, this disclosure contemplates forming any suitable edges **706** in any suitable manner.

(72) Vector Spaces and Embeddings

(73) FIG. **8** illustrates an example view of a vector space **800**. In particular embodiments, an object or an n-gram may be represented in a d-dimensional vector space, where d denotes any suitable number of dimensions. Although the vector space **800** is illustrated as a three-dimensional space, this is for illustrative purposes only, as the vector space **800** may be of any suitable dimension. In particular embodiments, an n-gram may be represented in the vector space **800** as a vector referred to as a term embedding. Each vector may comprise coordinates corresponding to a particular point in the vector space **800** (i.e., the terminal point of the vector). As an example and not by way of limitation, vectors **810**, **820**, and **830** may be represented as points in the vector space **800**, as illustrated in FIG. **8**. An n-gram may be mapped to a respective vector representation. As an example and not by way of limitation, n-grams t.sub.1 and t.sub.2 may be mapped to vectors custom character and custom character in the vector space **800**, respectively, by applying a function custom character defined by a dictionary, such that custom character= custom character(t.sub.1) and custom character=custom character(t.sub.2). As another example and not by way of limitation, a dictionary trained to map text to a vector representation may be utilized, or such a dictionary may be itself generated via training. As another example and not by way of limitation, a model, such as Word2vec, may be used to map an n-gram to a vector representation in the vector space **800**. In particular embodiments, an n-gram may be mapped to a vector representation in the vector space **800** by using a machine leaning model (e.g., a neural network). The machine learning model may have been trained using a sequence of training data (e.g., a corpus of objects each comprising n-grams).

(74) In particular embodiments, an object may be represented in the vector space **800** as a vector referred to as a feature vector or an object embedding. As an example and not by way of limitation, objects e.sub.1 and e.sub.2 may be mapped to vectors custom character and custom character in the vector space **800**, respectively, by applying a function custom character, such that custom character=custom character(e.sub.1) and custom character=custom character (e.sub.2). In particular embodiments, an object may be mapped to a vector based on one or more properties, attributes, or features of the object, relationships of the object with other objects, or any other suitable information associated with the object. As an example and not by way of limitation, a function custom character may map objects to vectors by feature extraction, which may start from an initial set of measured data and build derived values (e.g., features). As an example and not by way of limitation, an object comprising a video or an image may be mapped to a vector by using an algorithm to detect or isolate various desired portions or shapes of the object. Features used to calculate the vector may be based on information obtained from edge detection, corner detection, blob detection, ridge detection, scale-invariant feature transformation, edge direction, changing intensity, autocorrelation, motion detection, optical flow, thresholding, blob extraction, template matching, Hough transformation (e.g., lines, circles, ellipses, arbitrary shapes), or any other suitable information. As another example and not by way of limitation, an object comprising audio data may be mapped to a vector based on features such as a spectral slope, a tonality coefficient, an audio spectrum centroid, an audio spectrum envelope, a Mel-frequency cepstrum, or any other suitable information. In particular embodiments, when an object has data that is either too large to be efficiently processed or comprises redundant data, a function custom character may map the

object to a vector using a transformed reduced set of features (e.g., feature selection). In particular embodiments, a function custom character may map an object e to a vector custom character(e) based on one or more n-grams associated with object e. Although this disclosure describes representing an n-gram or an object in a vector space in a particular manner, this disclosure contemplates representing an n-gram or an object in a vector space in any suitable manner.

(75) In particular embodiments, the social-networking system **160** may calculate a similarity metric of vectors in vector space **800**. A similarity metric may be a cosine similarity, a Minkowski distance, a Mahalanobis distance, a Jaccard similarity coefficient, or any suitable similarity metric. As an example and not by way of limitation, a similarity metric of custom character and custom character may be a cosine similarity

(76) $\frac{.Math.\ v_1\ .Math.\ .Math.\ v_2\ .Math.}{.Math.\ .Math.\ v_1\ .Math.\ .Math.\ .Math.\ .Math.\ v_2\ .Math.\ .Math.}$.

As another example and not by way of limitation, a similarity metric of custom character and custom character may be a Euclidean distance ‖custom character−custom character‖. A similarity metric of two vectors may represent how similar the two objects or n-grams corresponding to the two vectors, respectively, are to one another, as measured by the distance between the two vectors in the vector space **800**. As an example and not by way of limitation, vector **810** and vector **820** may correspond to objects that are more similar to one another than the objects corresponding to vector **810** and vector **830**, based on the distance between the respective vectors. Although this disclosure describes calculating a similarity metric between vectors in a particular manner, this disclosure contemplates calculating a similarity metric between vectors in any suitable manner.

(77) More information on vector spaces, embeddings, feature vectors, and similarity metrics may be found in U.S. patent application Ser. No. 14/949,436, filed 23 Nov. 2015, U.S. patent application Ser. No. 15/286,315, filed 5 Oct. 2016, and U.S. patent application Ser. No. 15/365,789, filed 30 Nov. 2016, each of which is incorporated by reference.

(78) Artificial Neural Networks

(79) FIG. **9** illustrates an example artificial neural network ("ANN") **900**. In particular embodiments, an ANN may refer to a computational model comprising one or more nodes. Example ANN **900** may comprise an input layer **910**, hidden layers **920**, **930**, **960**, and an output layer **950**. Each layer of the ANN **900** may comprise one or more nodes, such as a node **905** or a node **915**. In particular embodiments, each node of an ANN may be connected to another node of the ANN. As an example and not by way of limitation, each node of the input layer **910** may be connected to one of more nodes of the hidden layer **920**. In particular embodiments, one or more nodes may be a bias node (e.g., a node in a layer that is not connected to and does not receive input from any node in a previous layer). In particular embodiments, each node in each layer may be connected to one or more nodes of a previous or subsequent layer. Although FIG. **9** depicts a particular ANN with a particular number of layers, a particular number of nodes, and particular connections between nodes, this disclosure contemplates any suitable ANN with any suitable number of layers, any suitable number of nodes, and any suitable connections between nodes. As an example and not by way of limitation, although FIG. **9** depicts a connection between each node of the input layer **910** and each node of the hidden layer **920**, one or more nodes of the input layer **910** may not be connected to one or more nodes of the hidden layer **920**.

(80) In particular embodiments, an ANN may be a feedforward ANN (e.g., an ANN with no cycles or loops where communication between nodes flows in one direction beginning with the input layer and proceeding to successive layers). As an example and not by way of limitation, the input to each node of the hidden layer **920** may comprise the output of one or more nodes of the input layer **910**. As another example and not by way of limitation, the input to each node of the output layer **950** may comprise the output of one or more nodes of the hidden layer **960**. In particular embodiments, an ANN may be a deep neural network (e.g., a neural network comprising at least two hidden

layers). In particular embodiments, an ANN may be a deep residual network. A deep residual network may be a feedforward ANN comprising hidden layers organized into residual blocks. The input into each residual block after the first residual block may be a function of the output of the previous residual block and the input of the previous residual block. As an example and not by way of limitation, the input into residual block N may be F(x)+x, where F(x) may be the output of residual block N−1, x may be the input into residual block N−1. Although this disclosure describes a particular ANN, this disclosure contemplates any suitable ANN.
(81) In particular embodiments, an activation function may correspond to each node of an ANN. An activation function of a node may define the output of a node for a given input. In particular embodiments, an input to a node may comprise a set of inputs. As an example and not by way of limitation, an activation function may be an identity function, a binary step function, a logistic function, or any other suitable function. As another example and not by way of limitation, an activation function for a node k may be the sigmoid function F.sub.k(s.sub.k)=
(82) $F_k(s_k) = \frac{1}{1 + e^{-s_k}}$,
the hyperbolic tangent function F.sub.k(s.sub.k)=
(83) $F_k(s_k) = \frac{e^{s_k} - e^{-s_k}}{e^{s_k} + e^{-s_k}}$,
the rectifier F.sub.k(s.sub.k)=max(0, s.sub.k), or any other suitable function F.sub.k(s.sub.k), where s.sub.k may be the effective input to node k. In particular embodiments, the input of an activation function corresponding to a node may be weighted. Each node may generate output using a corresponding activation function based on weighted inputs. In particular embodiments, each connection between nodes may be associated with a weight. As an example and not by way of limitation, a connection **925** between the node **905** and the node **915** may have a weighting coefficient of 0.4, which may indicate that 0.4 multiplied by the output of the node **905** is used as an input to the node **915**. As another example and not by way of limitation, the output y.sub.k of node k may be y.sub.k=F.sub.k(s.sub.k), where F.sub.k may be the activation function corresponding to node k, s.sub.k=Σ.sub.j(w.sub.jkx.sub.j) may be the effective input to node k, x.sub.j may be the output of a node j connected to node k, and w.sub.jk may be the weighting coefficient between node j and node k. In particular embodiments, the input to nodes of the input layer may be based on a vector representing an object. Although this disclosure describes particular inputs to and outputs of nodes, this disclosure contemplates any suitable inputs to and outputs of nodes. Moreover, although this disclosure may describe particular connections and weights between nodes, this disclosure contemplates any suitable connections and weights between nodes.
(84) In particular embodiments, an ANN may be trained using training data. As an example and not by way of limitation, training data may comprise inputs to the ANN **900** and an expected output. As another example and not by way of limitation, training data may comprise vectors each representing a training object and an expected label for each training object. In particular embodiments, training an ANN may comprise modifying the weights associated with the connections between nodes of the ANN by optimizing an objective function. As an example and not by way of limitation, a training method may be used (e.g., the conjugate gradient method, the gradient descent method, the stochastic gradient descent) to backpropagate the sum-of-squares error measured as a distances between each vector representing a training object (e.g., using a cost function that minimizes the sum-of-squares error). In particular embodiments, an ANN may be trained using a dropout technique. As an example and not by way of limitation, one or more nodes may be temporarily omitted (e.g., receive no input and generate no output) while training. For each training object, one or more nodes of the ANN may have some probability of being omitted. The nodes that are omitted for a particular training object may be different than the nodes omitted for other training objects (e.g., the nodes may be temporarily omitted on an object-by-object basis). Although this disclosure describes training an ANN in a particular manner, this disclosure contemplates training an ANN in any suitable manner.
(85) Privacy

(86) In particular embodiments, one or more objects (e.g., content or other types of objects) of a computing system may be associated with one or more privacy settings. The one or more objects may be stored on or otherwise associated with any suitable computing system or application, such as, for example, a social-networking system **160**, a client system **130**, an assistant system **140**, a third-party system **170**, a social-networking application, an assistant application, a messaging application, a photo-sharing application, or any other suitable computing system or application. Although the examples discussed herein are in the context of an online social network, these privacy settings may be applied to any other suitable computing system. Privacy settings (or “access settings”) for an object may be stored in any suitable manner, such as, for example, in association with the object, in an index on an authorization server, in another suitable manner, or any suitable combination thereof. A privacy setting for an object may specify how the object (or particular information associated with the object) can be accessed, stored, or otherwise used (e.g., viewed, shared, modified, copied, executed, surfaced, or identified) within the online social network. When privacy settings for an object allow a particular user or other entity to access that object, the object may be described as being “visible” with respect to that user or other entity. As an example and not by way of limitation, a user of the online social network may specify privacy settings for a user-profile page that identify a set of users that may access work-experience information on the user-profile page, thus excluding other users from accessing that information.
(87) In particular embodiments, privacy settings for an object may specify a “blocked list” of users or other entities that should not be allowed to access certain information associated with the object. In particular embodiments, the blocked list may include third-party entities. The blocked list may specify one or more users or entities for which an object is not visible. As an example and not by way of limitation, a user may specify a set of users who may not access photo albums associated with the user, thus excluding those users from accessing the photo albums (while also possibly allowing certain users not within the specified set of users to access the photo albums). In particular embodiments, privacy settings may be associated with particular social-graph elements. Privacy settings of a social-graph element, such as a node or an edge, may specify how the social-graph element, information associated with the social-graph element, or objects associated with the social-graph element can be accessed using the online social network. As an example and not by way of limitation, a particular concept node #**04** corresponding to a particular photo may have a privacy setting specifying that the photo may be accessed only by users tagged in the photo and friends of the users tagged in the photo. In particular embodiments, privacy settings may allow users to opt in to or opt out of having their content, information, or actions stored/logged by the social-networking system **160** or assistant system **140** or shared with other systems (e.g., a third-party system **170**). Although this disclosure describes using particular privacy settings in a particular manner, this disclosure contemplates using any suitable privacy settings in any suitable manner.
(88) In particular embodiments, privacy settings may be based on one or more nodes or edges of a social graph **700**. A privacy setting may be specified for one or more edges **706** or edge-types of the social graph **700**, or with respect to one or more nodes **702**, **704** or node-types of the social graph **700**. The privacy settings applied to a particular edge **706** connecting two nodes may control whether the relationship between the two entities corresponding to the nodes is visible to other users of the online social network. Similarly, the privacy settings applied to a particular node may control whether the user or concept corresponding to the node is visible to other users of the online social network. As an example and not by way of limitation, a first user may share an object to the social-networking system **160**. The object may be associated with a concept node **704** connected to a user node **702** of the first user by an edge **706**. The first user may specify privacy settings that apply to a particular edge **706** connecting to the concept node **704** of the object, or may specify privacy settings that apply to all edges **706** connecting to the concept node **704**. As another example and not by way of limitation, the first user may share a set of objects of a particular object-

type (e.g., a set of images). The first user may specify privacy settings with respect to all objects associated with the first user of that particular object-type as having a particular privacy setting (e.g., specifying that all images posted by the first user are visible only to friends of the first user and/or users tagged in the images).

(89) In particular embodiments, the social-networking system **160** may present a “privacy wizard” (e.g., within a webpage, a module, one or more dialog boxes, or any other suitable interface) to the first user to assist the first user in specifying one or more privacy settings. The privacy wizard may display instructions, suitable privacy-related information, current privacy settings, one or more input fields for accepting one or more inputs from the first user specifying a change or confirmation of privacy settings, or any suitable combination thereof. In particular embodiments, the social-networking system **160** may offer a “dashboard” functionality to the first user that may display, to the first user, current privacy settings of the first user. The dashboard functionality may be displayed to the first user at any appropriate time (e.g., following an input from the first user summoning the dashboard functionality, following the occurrence of a particular event or trigger action). The dashboard functionality may allow the first user to modify one or more of the first user's current privacy settings at any time, in any suitable manner (e.g., redirecting the first user to the privacy wizard).

(90) Privacy settings associated with an object may specify any suitable granularity of permitted access or denial of access. As an example and not by way of limitation, access or denial of access may be specified for particular users (e.g., only me, my roommates, my boss), users within a particular degree-of-separation (e.g., friends, friends-of-friends), user groups (e.g., the gaming club, my family), user networks (e.g., employees of particular employers, students or alumni of particular university), all users (“public”), no users (“private”), users of third-party systems **170**, particular applications (e.g., third-party applications, external websites), other suitable entities, or any suitable combination thereof. Although this disclosure describes particular granularities of permitted access or denial of access, this disclosure contemplates any suitable granularities of permitted access or denial of access.

(91) In particular embodiments, one or more servers **162** may be authorization/privacy servers for enforcing privacy settings. In response to a request from a user (or other entity) for a particular object stored in a data store **164**, the social-networking system **160** may send a request to the data store **164** for the object. The request may identify the user associated with the request and the object may be sent only to the user (or a client system **130** of the user) if the authorization server determines that the user is authorized to access the object based on the privacy settings associated with the object. If the requesting user is not authorized to access the object, the authorization server may prevent the requested object from being retrieved from the data store **164** or may prevent the requested object from being sent to the user. In the search-query context, an object may be provided as a search result only if the querying user is authorized to access the object, e.g., if the privacy settings for the object allow it to be surfaced to, discovered by, or otherwise visible to the querying user. In particular embodiments, an object may represent content that is visible to a user through a newsfeed of the user. As an example and not by way of limitation, one or more objects may be visible to a user's “Trending” page. In particular embodiments, an object may correspond to a particular user. The object may be content associated with the particular user, or may be the particular user's account or information stored on the social-networking system **160**, or other computing system. As an example and not by way of limitation, a first user may view one or more second users of an online social network through a “People You May Know” function of the online social network, or by viewing a list of friends of the first user. As an example and not by way of limitation, a first user may specify that they do not wish to see objects associated with a particular second user in their newsfeed or friends list. If the privacy settings for the object do not allow it to be surfaced to, discovered by, or visible to the user, the object may be excluded from the search results. Although this disclosure describes enforcing privacy settings in a particular manner, this

disclosure contemplates enforcing privacy settings in any suitable manner.
(92) In particular embodiments, different objects of the same type associated with a user may have different privacy settings. Different types of objects associated with a user may have different types of privacy settings. As an example and not by way of limitation, a first user may specify that the first user's status updates are public, but any images shared by the first user are visible only to the first user's friends on the online social network. As another example and not by way of limitation, a user may specify different privacy settings for different types of entities, such as individual users, friends-of-friends, followers, user groups, or corporate entities. As another example and not by way of limitation, a first user may specify a group of users that may view videos posted by the first user, while keeping the videos from being visible to the first user's employer. In particular embodiments, different privacy settings may be provided for different user groups or user demographics. As an example and not by way of limitation, a first user may specify that other users who attend the same university as the first user may view the first user's pictures, but that other users who are family members of the first user may not view those same pictures.
(93) In particular embodiments, the social-networking system **160** may provide one or more default privacy settings for each object of a particular object-type. A privacy setting for an object that is set to a default may be changed by a user associated with that object. As an example and not by way of limitation, all images posted by a first user may have a default privacy setting of being visible only to friends of the first user and, for a particular image, the first user may change the privacy setting for the image to be visible to friends and friends-of-friends.
(94) In particular embodiments, privacy settings may allow a first user to specify (e.g., by opting out, by not opting in) whether the social-networking system **160** or assistant system **140** may receive, collect, log, or store particular objects or information associated with the user for any purpose. In particular embodiments, privacy settings may allow the first user to specify whether particular applications or processes may access, store, or use particular objects or information associated with the user. The privacy settings may allow the first user to opt in or opt out of having objects or information accessed, stored, or used by specific applications or processes. The social-networking system **160** or assistant system **140** may access such information in order to provide a particular function or service to the first user, without the social-networking system **160** or assistant system **140** having access to that information for any other purposes. Before accessing, storing, or using such objects or information, the social-networking system **160** or assistant system **140** may prompt the user to provide privacy settings specifying which applications or processes, if any, may access, store, or use the object or information prior to allowing any such action. As an example and not by way of limitation, a first user may transmit a message to a second user via an application related to the online social network (e.g., a messaging app), and may specify privacy settings that such messages should not be stored by the social-networking system **160** or assistant system **140**.
(95) In particular embodiments, a user may specify whether particular types of objects or information associated with the first user may be accessed, stored, or used by the social-networking system **160** or assistant system **140**. As an example and not by way of limitation, the first user may specify that images sent by the first user through the social-networking system **160** or assistant system **140** may not be stored by the social-networking system **160** or assistant system **140**. As another example and not by way of limitation, a first user may specify that messages sent from the first user to a particular second user may not be stored by the social-networking system **160** or assistant system **140**. As yet another example and not by way of limitation, a first user may specify that all objects sent via a particular application may be saved by the social-networking system **160** or assistant system **140**.
(96) In particular embodiments, privacy settings may allow a first user to specify whether particular objects or information associated with the first user may be accessed from particular client systems **130** or third-party systems **170**. The privacy settings may allow the first user to opt in or opt out of having objects or information accessed from a particular device (e.g., the phone book on a user's

smart phone), from a particular application (e.g., a messaging app), or from a particular system (e.g., an email server). The social-networking system **160** or assistant system **140** may provide default privacy settings with respect to each device, system, or application, and/or the first user may be prompted to specify a particular privacy setting for each context. As an example and not by way of limitation, the first user may utilize a location-services feature of the social-networking system **160** or assistant system **140** to provide recommendations for restaurants or other places in proximity to the user. The first user's default privacy settings may specify that the social-networking system **160** or assistant system **140** may use location information provided from a client device **130** of the first user to provide the location-based services, but that the social-networking system **160** or assistant system **140** may not store the location information of the first user or provide it to any third-party system **170**. The first user may then update the privacy settings to allow location information to be used by a third-party image-sharing application in order to geo-tag photos.

(97) Privacy Settings Based on Location

(98) In particular embodiments, privacy settings may allow a user to specify one or more geographic locations from which objects can be accessed. Access or denial of access to the objects may depend on the geographic location of a user who is attempting to access the objects. As an example and not by way of limitation, a user may share an object and specify that only users in the same city may access or view the object. As another example and not by way of limitation, a first user may share an object and specify that the object is visible to second users only while the first user is in a particular location. If the first user leaves the particular location, the object may no longer be visible to the second users. As another example and not by way of limitation, a first user may specify that an object is visible only to second users within a threshold distance from the first user. If the first user subsequently changes location, the original second users with access to the object may lose access, while a new group of second users may gain access as they come within the threshold distance of the first user.

(99) Privacy Settings for User-Authentication and Experience-Personalization Information

(100) In particular embodiments, the social-networking system **160** or assistant system **140** may have functionalities that may use, as inputs, personal or biometric information of a user for user-authentication or experience-personalization purposes. A user may opt to make use of these functionalities to enhance their experience on the online social network. As an example and not by way of limitation, a user may provide personal or biometric information to the social-networking system **160** or assistant system **140**. The user's privacy settings may specify that such information may be used only for particular processes, such as authentication, and further specify that such information may not be shared with any third-party system **170** or used for other processes or applications associated with the social-networking system **160** or assistant system **140**. As another example and not by way of limitation, the social-networking system **160** may provide a functionality for a user to provide voice-print recordings to the online social network. As an example and not by way of limitation, if a user wishes to utilize this function of the online social network, the user may provide a voice recording of his or her own voice to provide a status update on the online social network. The recording of the voice-input may be compared to a voice print of the user to determine what words were spoken by the user. The user's privacy setting may specify that such voice recording may be used only for voice-input purposes (e.g., to authenticate the user, to send voice messages, to improve voice recognition in order to use voice-operated features of the online social network), and further specify that such voice recording may not be shared with any third-party system **170** or used by other processes or applications associated with the social-networking system **160**. As another example and not by way of limitation, the social-networking system **160** may provide a functionality for a user to provide a reference image (e.g., a facial profile, a retinal scan) to the online social network. The online social network may compare the reference image against a later-received image input (e.g., to authenticate the user, to tag the user in

photos). The user's privacy setting may specify that such voice recording may be used only for a limited purpose (e.g., authentication, tagging the user in photos), and further specify that such voice recording may not be shared with any third-party system **170** or used by other processes or applications associated with the social-networking system **160**.

(101) Systems and Methods

(102) FIG. **10** illustrates an example computer system **1000**. In particular embodiments, one or more computer systems **1000** perform one or more steps of one or more methods described or illustrated herein. In particular embodiments, one or more computer systems **1000** provide functionality described or illustrated herein. In particular embodiments, software running on one or more computer systems **1000** performs one or more steps of one or more methods described or illustrated herein or provides functionality described or illustrated herein. Particular embodiments include one or more portions of one or more computer systems **1000**. Herein, reference to a computer system may encompass a computing device, and vice versa, where appropriate. Moreover, reference to a computer system may encompass one or more computer systems, where appropriate.

(103) This disclosure contemplates any suitable number of computer systems **1000**. This disclosure contemplates computer system **1000** taking any suitable physical form. As example and not by way of limitation, computer system **1000** may be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC) (such as, for example, a computer-on-module (COM) or system-on-module (SOM)), a desktop computer system, a laptop or notebook computer system, an interactive kiosk, a mainframe, a mesh of computer systems, a mobile telephone, a personal digital assistant (PDA), a server, a tablet computer system, or a combination of two or more of these. Where appropriate, computer system **1000** may include one or more computer systems **1000**; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **1000** may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example and not by way of limitation, one or more computer systems **1000** may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems **1000** may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

(104) In particular embodiments, computer system **1000** includes a processor **1002**, memory **1004**, storage **1006**, an input/output (I/O) interface **1008**, a communication interface **1010**, and a bus **1012**. Although this disclosure describes and illustrates a particular computer system having a particular number of particular components in a particular arrangement, this disclosure contemplates any suitable computer system having any suitable number of any suitable components in any suitable arrangement.

(105) In particular embodiments, processor **1002** includes hardware for executing instructions, such as those making up a computer program. As an example and not by way of limitation, to execute instructions, processor **1002** may retrieve (or fetch) the instructions from an internal register, an internal cache, memory **1004**, or storage **1006**; decode and execute them; and then write one or more results to an internal register, an internal cache, memory **1004**, or storage **1006**. In particular embodiments, processor **1002** may include one or more internal caches for data, instructions, or addresses. This disclosure contemplates processor **1002** including any suitable number of any suitable internal caches, where appropriate. As an example and not by way of limitation, processor **1002** may include one or more instruction caches, one or more data caches, and one or more translation lookaside buffers (TLBs). Instructions in the instruction caches may be copies of instructions in memory **1004** or storage **1006**, and the instruction caches may speed up retrieval of those instructions by processor **1002**. Data in the data caches may be copies of data in memory

1004 or storage **1006** for instructions executing at processor **1002** to operate on; the results of previous instructions executed at processor **1002** for access by subsequent instructions executing at processor **1002** or for writing to memory **1004** or storage **1006**; or other suitable data. The data caches may speed up read or write operations by processor **1002**. The TLBs may speed up virtual-address translation for processor **1002**. In particular embodiments, processor **1002** may include one or more internal registers for data, instructions, or addresses. This disclosure contemplates processor **1002** including any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor **1002** may include one or more arithmetic logic units (ALUs); be a multi-core processor; or include one or more processors **1002**. Although this disclosure describes and illustrates a particular processor, this disclosure contemplates any suitable processor.

(106) In particular embodiments, memory **1004** includes main memory for storing instructions for processor **1002** to execute or data for processor **1002** to operate on. As an example and not by way of limitation, computer system **1000** may load instructions from storage **1006** or another source (such as, for example, another computer system **1000**) to memory **1004**. Processor **1002** may then load the instructions from memory **1004** to an internal register or internal cache. To execute the instructions, processor **1002** may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor **1002** may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor **1002** may then write one or more of those results to memory **1004**. In particular embodiments, processor **1002** executes only instructions in one or more internal registers or internal caches or in memory **1004** (as opposed to storage **1006** or elsewhere) and operates only on data in one or more internal registers or internal caches or in memory **1004** (as opposed to storage **1006** or elsewhere). One or more memory buses (which may each include an address bus and a data bus) may couple processor **1002** to memory **1004**. Bus **1012** may include one or more memory buses, as described below. In particular embodiments, one or more memory management units (MMUs) reside between processor **1002** and memory **1004** and facilitate accesses to memory **1004** requested by processor **1002**. In particular embodiments, memory **1004** includes random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Moreover, where appropriate, this RAM may be single-ported or multi-ported RAM. This disclosure contemplates any suitable RAM. Memory **1004** may include one or more memories **1004**, where appropriate. Although this disclosure describes and illustrates particular memory, this disclosure contemplates any suitable memory.

(107) In particular embodiments, storage **1006** includes mass storage for data or instructions. As an example and not by way of limitation, storage **1006** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **1006** may include removable or non-removable (or fixed) media, where appropriate. Storage **1006** may be internal or external to computer system **1000**, where appropriate. In particular embodiments, storage **1006** is non-volatile, solid-state memory. In particular embodiments, storage **1006** includes read-only memory (ROM). Where appropriate, this ROM may be mask-programmed ROM, programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), electrically alterable ROM (EAROM), or flash memory or a combination of two or more of these. This disclosure contemplates mass storage **1006** taking any suitable physical form. Storage **1006** may include one or more storage control units facilitating communication between processor **1002** and storage **1006**, where appropriate. Where appropriate, storage **1006** may include one or more storages **1006**. Although this disclosure describes and illustrates particular storage, this disclosure contemplates any suitable storage.

(108) In particular embodiments, I/O interface **1008** includes hardware, software, or both, providing one or more interfaces for communication between computer system **1000** and one or

more I/O devices. Computer system **1000** may include one or more of these I/O devices, where appropriate. One or more of these I/O devices may enable communication between a person and computer system **1000**. As an example and not by way of limitation, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, still camera, stylus, tablet, touch screen, trackball, video camera, another suitable I/O device or a combination of two or more of these. An I/O device may include one or more sensors. This disclosure contemplates any suitable I/O devices and any suitable I/O interfaces **1008** for them. Where appropriate, I/O interface **1008** may include one or more device or software drivers enabling processor **1002** to drive one or more of these I/O devices. I/O interface **1008** may include one or more I/O interfaces **1008**, where appropriate. Although this disclosure describes and illustrates a particular I/O interface, this disclosure contemplates any suitable I/O interface.

(109) In particular embodiments, communication interface **1010** includes hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **1000** and one or more other computer systems **1000** or one or more networks. As an example and not by way of limitation, communication interface **1010** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC) or wireless adapter for communicating with a wireless network, such as a WI-FI network. This disclosure contemplates any suitable network and any suitable communication interface **1010** for it. As an example and not by way of limitation, computer system **1000** may communicate with an ad hoc network, a personal area network (PAN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **1000** may communicate with a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a WI-MAX network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network), or other suitable wireless network or a combination of two or more of these. Computer system **1000** may include any suitable communication interface **1010** for any of these networks, where appropriate. Communication interface **1010** may include one or more communication interfaces **1010**, where appropriate. Although this disclosure describes and illustrates a particular communication interface, this disclosure contemplates any suitable communication interface.

(110) In particular embodiments, bus **1012** includes hardware, software, or both coupling components of computer system **1000** to each other. As an example and not by way of limitation, bus **1012** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCIe) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or another suitable bus or a combination of two or more of these. Bus **1012** may include one or more buses **1012**, where appropriate. Although this disclosure describes and illustrates a particular bus, this disclosure contemplates any suitable bus or interconnect.

(111) Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs, optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, solid-state drives (SSDs), RAM-drives, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a

combination of volatile and non-volatile, where appropriate.
(112) Miscellaneous
(113) Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.
(114) The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, feature, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Additionally, although this disclosure describes or illustrates particular embodiments as providing particular advantages, particular embodiments may provide none, some, or all of these advantages.

Claims

1. A method comprising, by a client system: receiving, at the client system, a speech input from a user and a visual input captured by one or more cameras of the client system, wherein the speech input comprises a textual input spoken by the user, and wherein the visual input depicts a real-time view captured by the one or more cameras, wherein the real-time view comprises one or more visual concepts and one or more attributes associated with the one or more visual concepts, and wherein the textual input comprises a co-reference to one or more of the visual concepts; resolving, based on the one or more attributes and the co-reference, one or more entities corresponding to the one or more visual concepts associated with the co-reference; and presenting, at the client system, a communication content responsive to the speech input and the visual input, wherein the communication content comprises information associated with executing results of one or more tasks corresponding to the one or more resolved entities.
2. The method of claim 1, wherein the speech input comprises one or more audio clips, and wherein the visual input comprises one or more of an image or a video clip.
3. The method of claim 1, further comprising: determining, based on the visual input, the one or more visual concepts and the one or more attributes associated with the one or more visual concepts.
4. The method of claim 3, wherein determining the one or more visual concepts and the one or more attributes is based on one or more machine-learning models comprising one or more of: a support vector machine; a regression model; or a convolutional neural network.
5. The method of claim 1, wherein the one or more visual concepts comprise one or more of a person, a location, a business, or an object.
6. The method of claim 5, wherein the one or more visual concepts comprise one or more people, and wherein the method further comprises: determining the one or more people based on facial

recognition.
7. The method of claim 5, wherein the one or more visual concepts comprise one or more objects, and wherein the method further comprises: determining the one or more objects based on object detection.
8. The method of claim 1, further comprising generating a feature representation for the visual input.
9. The method of claim 1, further comprising identifying one or more intents and one or more slots based on one or more of the speech input or the visual input.
10. The method of claim 9, further comprising executing the one or more tasks based on the identified intents and slots.
11. The method of claim 1, wherein the communication content comprises one or more of: a character string; an audio clip; an image; or a video clip.
12. The method of claim 1, further comprising determining one or more modalities for the communication content.
13. The method of claim 12, wherein determining the one or more modalities for the communication content comprises: identifying contextual information associated with a user of the client system; identifying contextual information associated with the client system; and determining the one or more modalities based on the contextual information associated with the user and the contextual information associated with the client system.
14. The method of claim 1, further comprising: generating a plurality of tasks based on the visual input; and receiving, at the client system, a user selection of the one or more tasks from the plurality of tasks by a user of the client system.
15. The method of claim 1, further comprising storing the one or more visual concepts in a dialog state.
16. The method of claim 1, further comprising: generating, based on the visual input, a media content object portraying the one or more visual concepts; receiving, at the client system, a user interaction with the media content object; and executing one or more additional tasks responsive to the user interaction with the media content object.
17. One or more computer-readable non-transitory non- volatile storage media embodying software that is operable when executed by a client system to: receive, at a client system, a speech input from a user and a visual input captured by one or more cameras of the client system, wherein the speech input comprises a textual input spoken by the user, and wherein the visual input depicts a real-time view captured by the one or more cameras, wherein the real-time view comprises one or more visual concepts and one or more attributes associated with the one or more visual concepts, and wherein the textual input comprises a co-reference to one or more of the visual concepts; resolve, based on the one or more attributes and the co-reference, one or more entities corresponding to the one or more visual concepts associated with the co-reference; and present, at the client system, a communication content responsive to the speech input and the visual input, wherein the communication content comprises information associated with executing results of one or more tasks corresponding to the one or more resolved entities.
18. A client system comprising: one or more processors; and a non-transitory non-volatile memory coupled to the processors comprising instructions executable by the processors, the processors operable when executing the instructions to: receive, at a client system, a speech input from a user and a visual input captured by one or more cameras of the client system, wherein the speech input comprises a textual input spoken by the user, and wherein the visual input depicts a real-time view captured by the one or more cameras, wherein the real-time view comprises one or more visual concepts and one or more attributes associated with the one or more visual concepts, and wherein the textual input comprises a co-reference to one or more of the visual concepts; resolve, based on the one or more attributes and the co-reference, one or more entities corresponding to the one or more visual concepts associated with the co-reference; and present, at the client system, a

communication content responsive to the speech input and the visual input, wherein the communication content comprises information associated with executing results of one or more tasks corresponding to the one or more resolved entities.